GEK-SA-1056

Boiler Stress Monitoring

User Guide



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Safety Symbol Legend



Warning

Indicates a procedure, condition, or statement that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure, condition, or statement that should be strictly followed to improve these applications.

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Preface

This document provides the low level operation and basic configuration of the GE Boiler Stress Monitoring product. Interaction with the product and use of the product outputs are also discussed.

The Boiler Stress Monitoring (BSM) product allows maintenance and operating personnel to manage Heat Recovery Steam Generator (HRSG) operation. It should not be used as a replacement to the recommendations and limits set forth in the operational and maintenance instructions for the equipment provided. The BSM product should not replace the proper inspection and maintenance of HRSG equipment by properly trained, qualified operators and maintenance personnel.

This manual does not provide any guidance on interpretation of stress and life consumption data. All accumulated life data should be considered relative and not absolute. Consult a qualified expert for the proper interpretation and use of the data. GE assumes no liability with respect to accuracy of the data reported against useful life of the equipment provided beyond the obligations set forth in the contracts.

Terms

Parameter or Keyword Units	Description
Creep Damage	Long term exposure to high temperature
Creep Stress	Time-dependent deformation at elevated temperature and constant stress
Danish Xergy Technology (DXT)	Supplier of the EN 12952 compliant calculation engine
EN12952-4	European Standard for Water-Tube Boilers and Auxiliary Installations Part 4: In-Service Boiler life expectancy calculations
Ethernet Global Data (EGD)	Producer/consumer protocol that provides data exchange between nodes at a nominal frame rate of 25 Hz
Fatigue Damage	Structural damage due to exceeding yield strength
ICS	Integrated Control System
Mechanical Stress	Stress due to internal pressures of the various shell and pressure vessels which make up the HRSG
Thermal Stress	Thermal gradients due to thermal transients across thick wall components of the HRSG
Total Damage	Summation of creep and fatigue damage
Unit Data Highway (UDH)	Control layer EGD network for per-to-peer communications

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1 System Overview

The BSM product is applicable to a wide range of boiler types with varying component geometries and metallurgy and is compatible with all industry standard pressure part design codes.

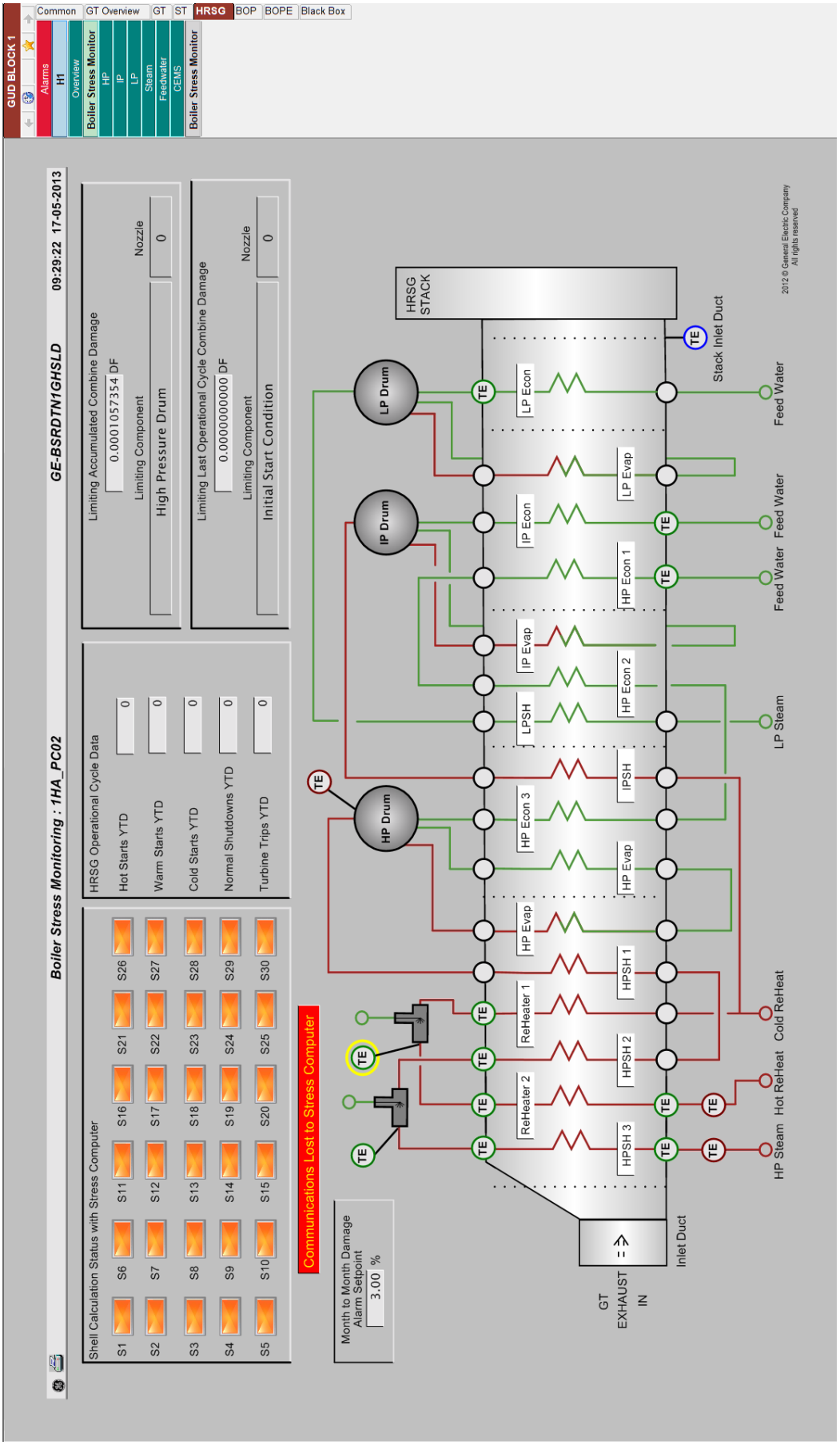
GE has developed the BSM system as a fully integrated component in the plant Integrated Control System (ICS) to improve the plant operating process and assist in the management of plant assets. The BSM system provides online calculation of thermal, mechanical and creep stress, and corresponding fatigue and creep fractions in steam generator components. This enables plant personnel to evaluate heat recovery steam generator (HRSG) component life consumption against plant operation in a consistent analysis methodology.

The BSM software applies steady-state methodology of the European Norm standards for the design and dimensioning of shell boilers, with enhancements to allow online, real-time calculations. Coupled with the distributed control system (DCS) and Historian, the monitored component stress levels and fatigue and/or creep damage fractions can be archived for later processing against various plant process data.

Typical monitoring components include:

- High Pressure Drum
- High Pressure Super Heater Outlet Headers
- High Pressure Manifolds
- High Pressure Super Heater Inlet Headers
- Reheater Outlet Headers
- Reheater Manifolds
- High Pressure Economizer Inlet Header
- High Pressure Super Heater Attemperator Discharge Pipe
- Reheater Attemperator Discharge Pipe

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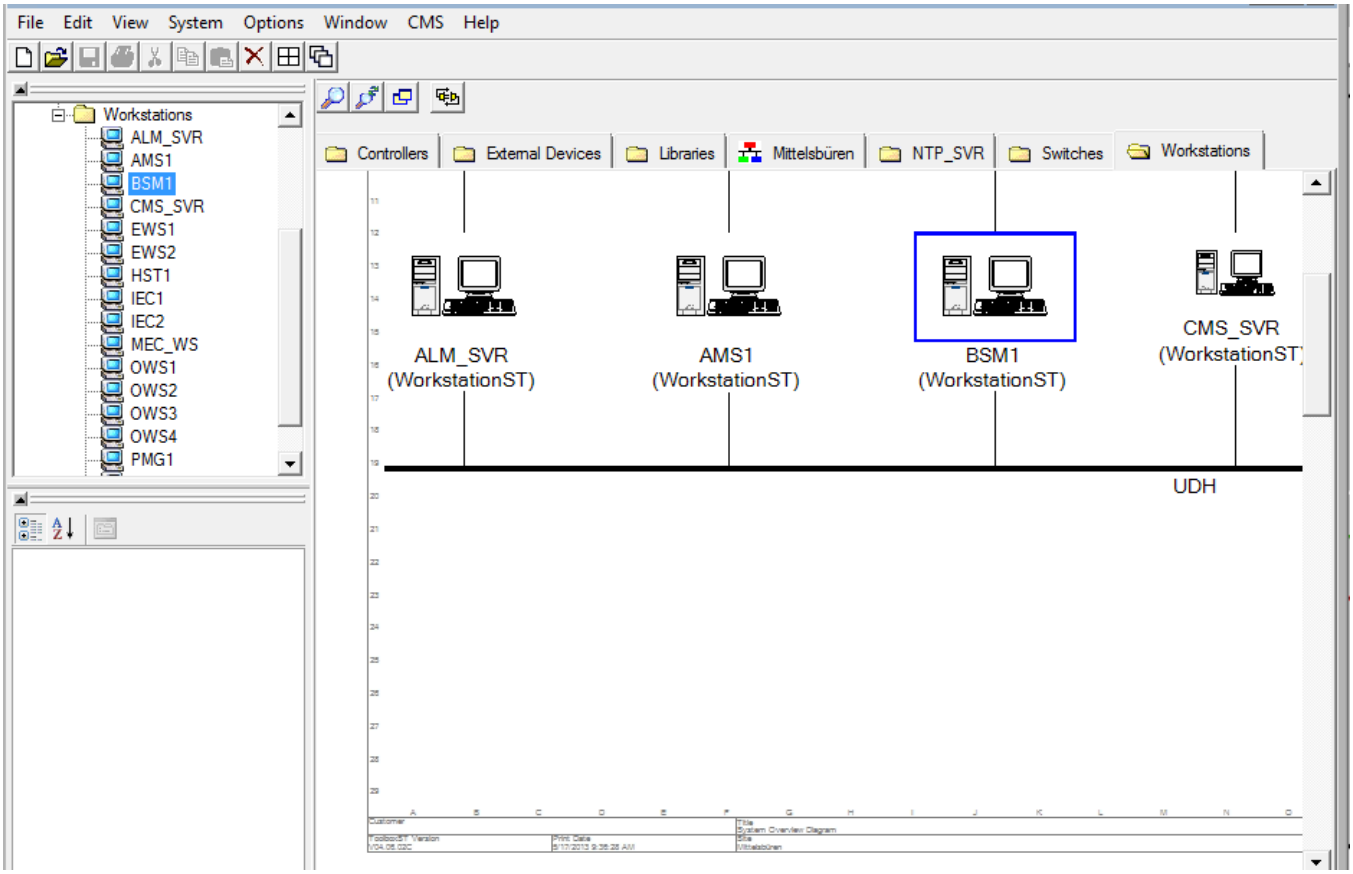
Typical BSM Screen

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The BSM product includes:

- A dedicated computer workstation, *BSM1*, that is a component of the plant ICS. This workstation has both the WorkStationST* application and the Danish Xergy Technology (DXT) calculation engine. The calculation engine performs the actual stress, fatigue, and creep calculations. The calculation engine interfaces with the Mark* VIe control system through WorkStationST features for the exchange of process and calculated variables. The WorkStationST application allows the workstation to communicate and exchange data with the Mark VIe control system and plant data historian.
- Dedicated thermocouples to measure mid-wall and inner-wall temperature for each component (drum or shell) being monitored. The thermocouples are located in strategic areas of the component, as identified by the HRSG supplier. The temperature gradient in the component, coupled with the temperature rate of change, is critical in calculating the thermal and creep stress. Thermocouple signal conditioning includes conversion to a 4 –20 mA analog value before being passed as an input to the ICS controller.
- A pressure signal for each component is required for both stress and fatigue calculations.
- A Balance of Plant (BOP) controller for each HRSG manages the data exchange with the BSM workstation. This controller serves as the accumulator for the calculated values from the BSM calculations. It also performs diagnostic monitoring and alarming, as well as archiving to the plant data historian.
- The plant data historian is the repository for the accumulated values from each monitored component. This history is a critical data source for analyzing cyclic effects on the HRSG.
- DXT application software adheres to EN 12952 standards and uses rain flow algorithms to assess stress and damage in relation to load cycles. It runs on the dedicated workstation, BSM1.

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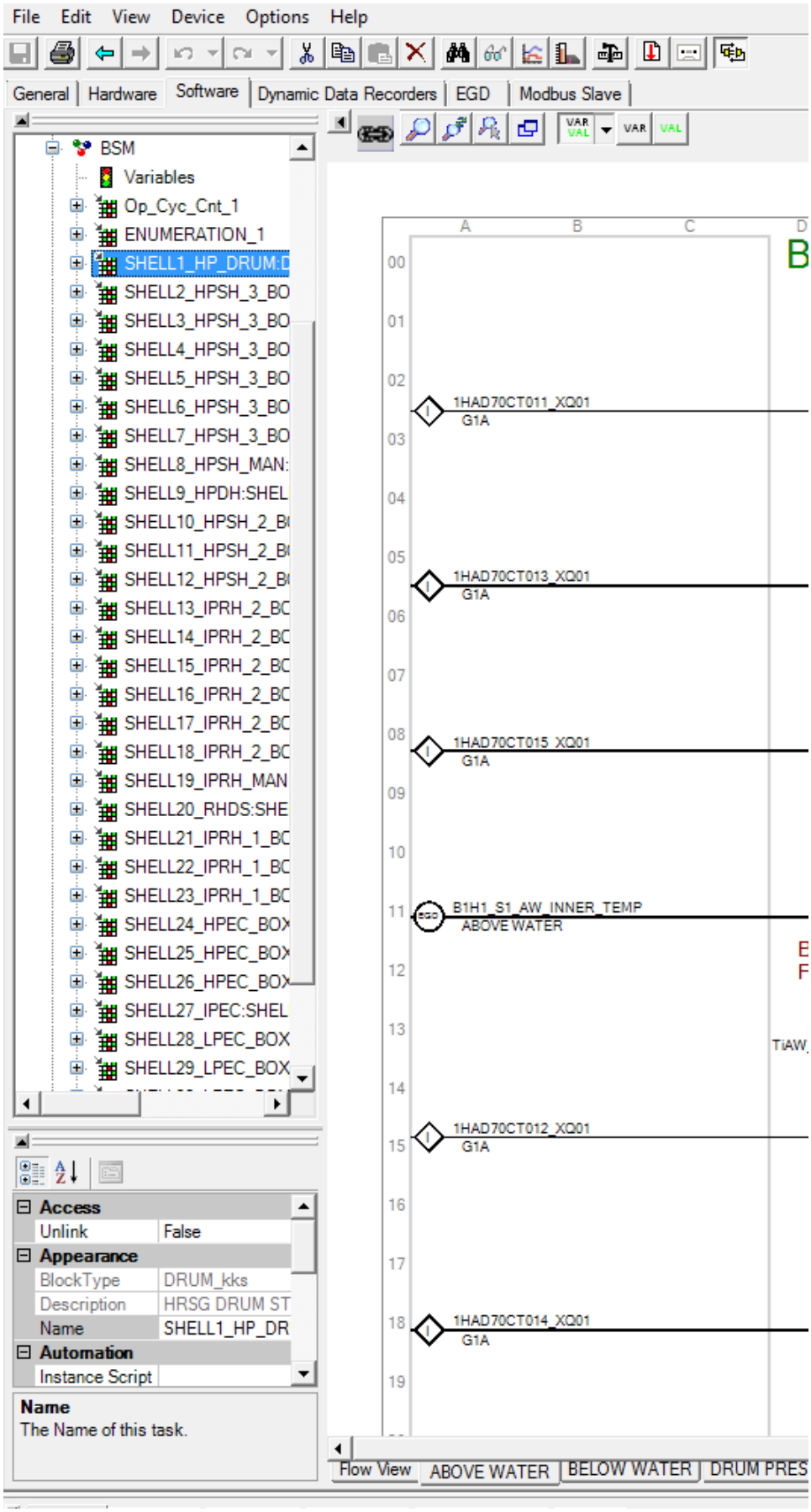


BSM1 Component in a Mark VIe Control System

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Example of Component Metal Thermocouples Connecting to the Control System as Standard I/O

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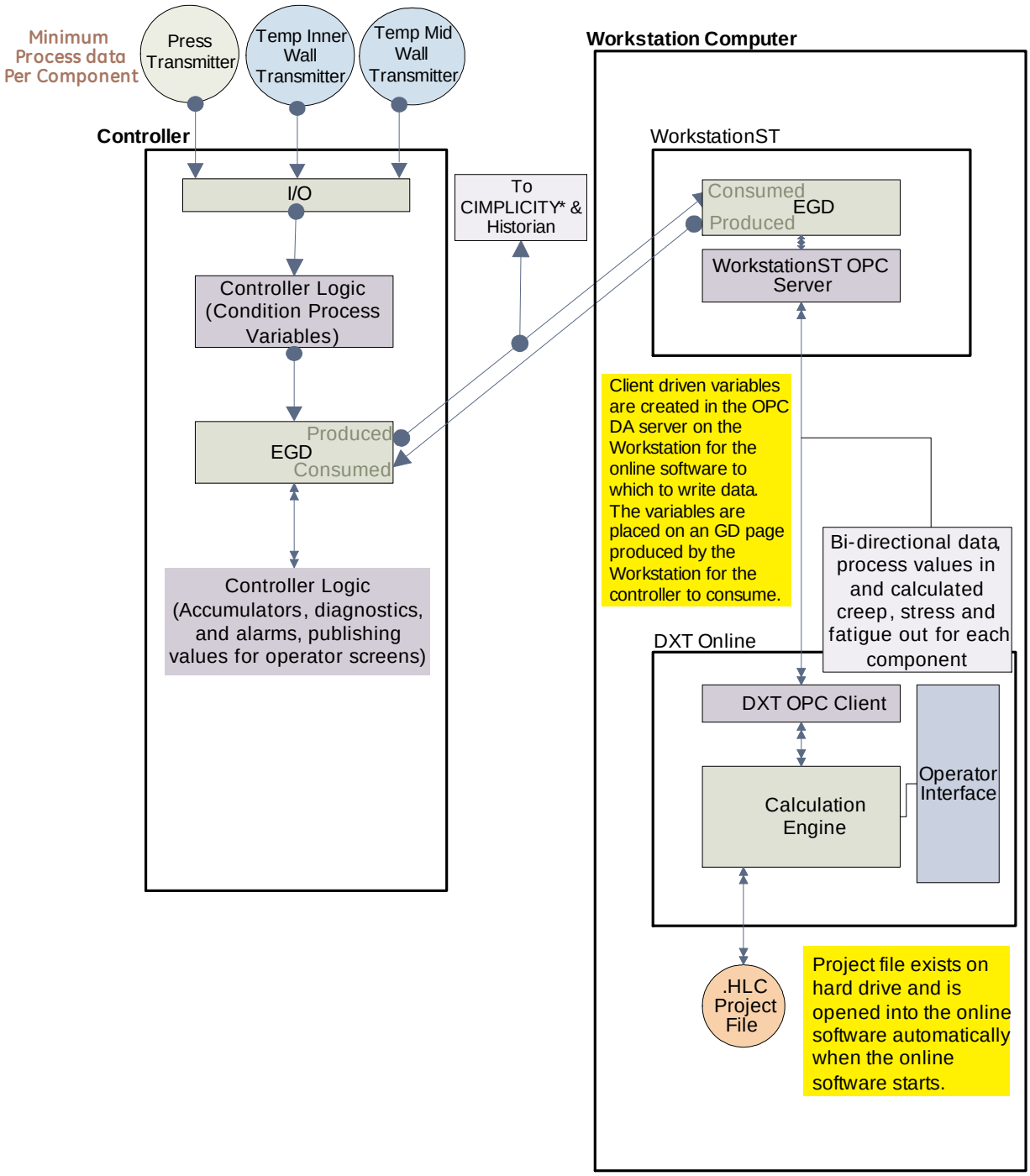


Example of Location of Controller Logic Inside the HRSG Controller

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2 System Communication

The BSM product communication uses various standard Mark VIe communication protocols, supporting communication between the controller and workstation, inner-workstation communication with the DXT online calculation engine, Human-machine Interface (HMI) (operator control interface) screens, and the data plant historian.



Simplified BSM System Communications

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2.1 Communication Between the Controllers and Workstation

Refer to GEH-6721, Volume I, Mark VIe Control System Guide.

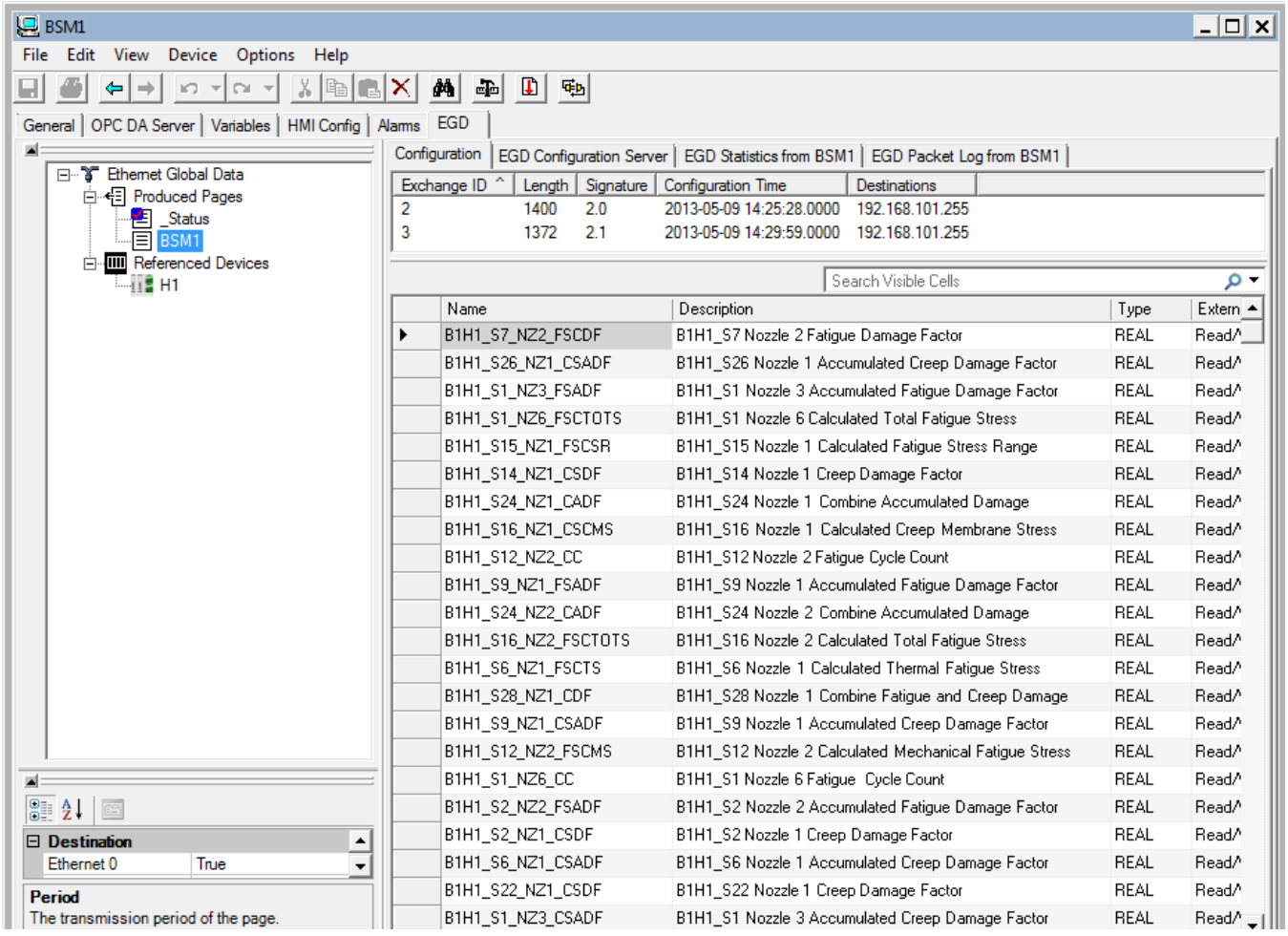
HRSG controllers running BSM software are required to reference the BSM1 workstation on the EGD to consume the required variables.

Bi-directional data flow between the individual HRSG controllers and the BSM1 workstation uses Ethernet Global Data (EGD) protocol for efficient data packet exchange. Data transfer uses the controller Unit Data Highway (UDH) and requires the BSM1 workstation to reference each of the block and HRSG controllers. The workstation consumes the data published by each controller to collect the process data and publish on the BSM1 workstation OPC-DA Server for request by the DXT calculation engine OPC-DA client. The data generated by the DXT calculation engine is passed to the controllers by the WorkstationST OPC-DA function for Client Driven variables. The variables to which the DXT online calculation engine writes are created on the BSM1 workstation and assigned to a dedicated EGD page named BSM1.



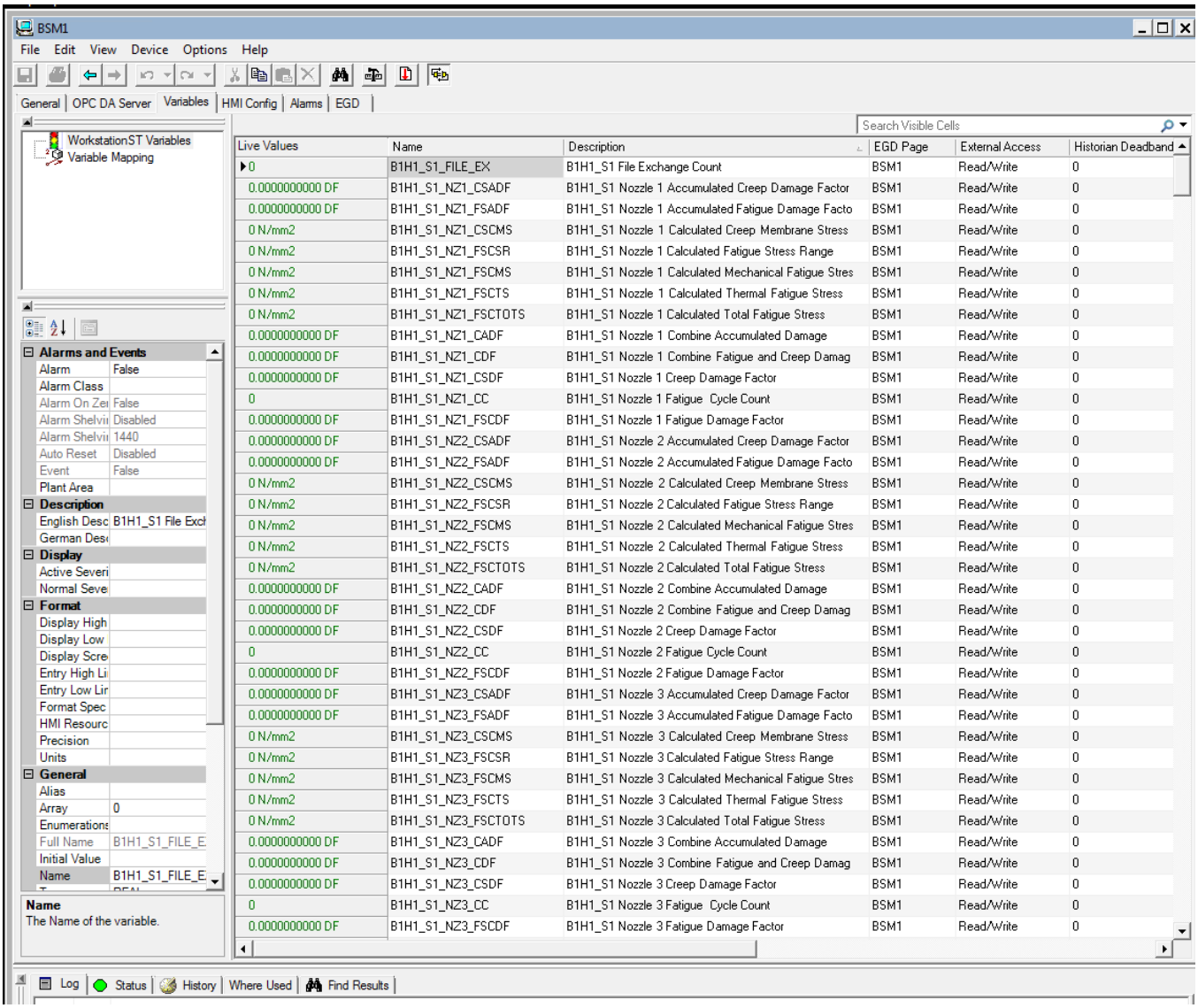
Caution

BSM1 workstation variables should never be modified. The variable format is critical to the accumulation of creep and fatigue in the controllers and the storage of BSM data to the plant data historian.



Referenced Controllers and BSM1 EGD Page for BSM1 Workstation

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Example of Client Driven Variables on BSM1 Workstation

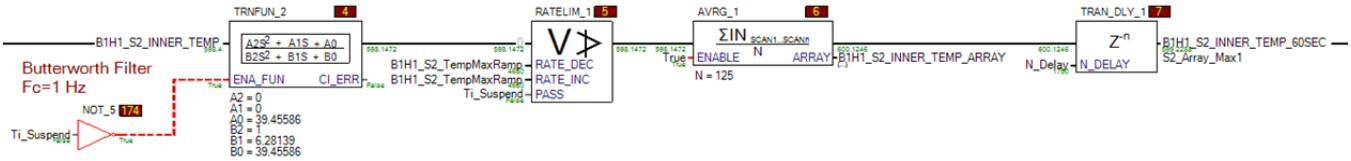
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2.2 Data Transfer Between the Controller and DXT Calculation Engine

The metal temperature and vessel pressure signals originate as 4–20 mA analog signals from a field-mounted transmitter and are connected to Mark VIe ICS inputs. The controller receiving the signals performs filtering and data quality testing before passing the signals through the workstation to the DXT online calculation engine.

The filtering component comprises a Butterworth™ filter with a 1 Hz cutoff frequency (fc) to filter out high frequency noise. The Butterworth filter is followed by a rate limiter that has the rate cutoff above the maximum rate that the actual process can change. The rate limiter helps to filter the zone above real data and under 1 Hz. The rate limiter is followed by a 10 second rolling average operating at frame rate. At frame rate, the 10 second frame rate is a rolling average of 125 samples (10 seconds/0.08 second frame rate). The rolling average helps filter out low level sinusoidal noise.

The final filtering is the passing of the signals through a data buffer to delay the signal by 60 seconds before being published to the workstation. The 60 second delay allows more than adequate time for the ICS to detect data quality issues and send the appropriate calculation command to the DXT online calculation engine to ensure no calculations are performed with bad quality data.



Standard Process Signal Filtering Scheme

Data quality testing includes checking for bad quality status as detected by the input circuitry of the ICS, metal temperatures greater than the maximum scaled 4–20 mA of the transmitter or the calculated temperature maximum of the metal in the component as calculated by DXT for the online calculation engine, temperature below 5°C (41 °F), unless the HRSG is offline, then the temperature setpoint moves to 0°C (32 °F). When a bad quality status is detected on either of the metal temperatures or pressure for a component, the suspend variable originating in the ICS is appropriately updated to signal the online calculation engine to suspend calculations that use the signals tagged as bad quality. A suspend variable exists for each component monitored on each HRSG. The suspend variables have the format *BxHx_Sx_Suspend*.

The data output from the DXT online calculation engine includes the following important variables, which are calculated and updated every 10 seconds:

- FILE_EX — Verifies that communications are active and provides feedback that the online software has received and responded to a GT halt or signal bad quality calculation suspend command.
- BxHx_Sx_NZx_CSCMS — Calculated Creep Membrane Stress
- BxHx_Sx_NZx_CSDF — Creep Damage Factor
- BxHx_Sx_NZx_FSCDF — Fatigue Damage Factor
- BxHx_Sx_NZx_FSCMS — Calculated Mechanical Fatigue Stress
- BxHx_Sx_NZx_FSCTS — Calculated Thermal Fatigue Stress

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Original Fill Exchange Counts - Superceded by Matrix Below												
			File Exchange Count									
Tref	Creep	Fatigue	Min	Max								
x	x	x	1	1000000								
o	o	o	-1	-9								
x	o	x	-10	-19								
x	x	o	-20	-29								
o	x	x	-30	-39								
o	o	x	-40	-49								
o	x	o	-50	-59								
x	o	o	-60	-69								
Tref - Refernce Temperature		x	Parameter valid									
		o	Parameter not valid									
FILE_EX Variable Feedback Matrix to DCS.												
					Bad signal on							
GT-status	Suspend	File exchange count.		GT running	Ti	Tm	TiAW	TmAW	TiBW	TmBW	P	
1	0	1 to 100000		Yes	NA	NA	NA	NA	NA	NA	NA	
0	0 to 7	-70 to -79		No	NA	NA	NA	NA	NA	NA	NA	
1	1	-80 to -89		Yes	0	x	x	x	x	x	x	
1	2	90 to -99		Yes	x	0	x	x	x	x	x	
1	3	-100 to -109		Yes	x	x	0	x	x	x	x	
1	4	-110 to -119		Yes	x	x	x	0	x	x	x	
1	5	120 to -129		Yes	x	x	x	x	0	x	x	
1	6	-130 to -139		Yes	x	x	x	x	x	0	x	
1	7	140 to -149		Yes	x	x	x	x	x	x	0	
Count recycles the range as long as the condition exists												
					Ti - Temperature Inner				x	Parameter valid		
					Tm - Temperature Middle				o	Parameter not valid		
				TiAW - Temperature Inner Above Water								
				TmAW -Temperature Middle Above Water								
				TiBW - Temperature Inner Below Water								
				TmBW -Temperature Middle Below Water								
				P - Pressure								

FILE_EX Variable Possible Values

A two hour delay allows for uniform cooling of the components of the HRSG.

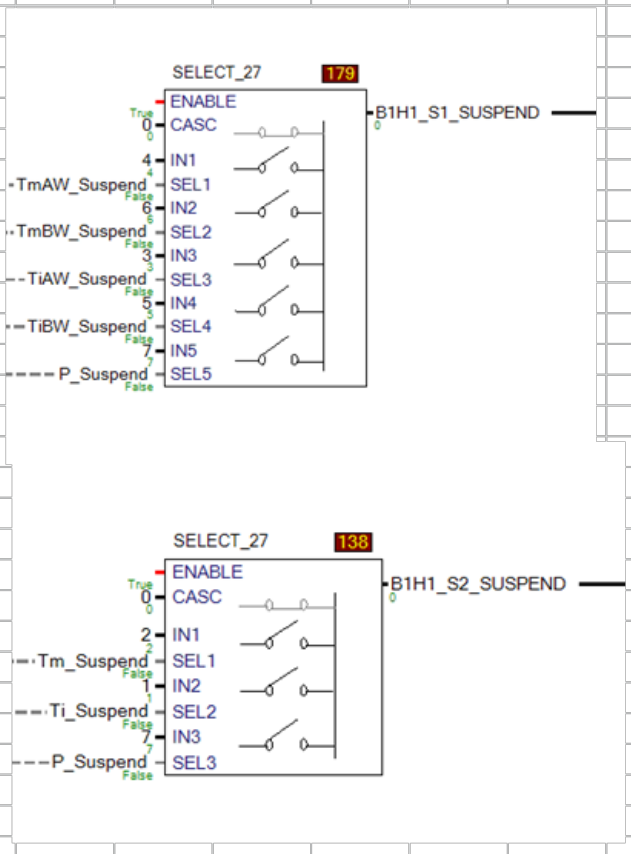
The GT status variable takes the format, *BxHx_GT_STATUS*, and indicates that the gas turbine for the particular HRSG is either firing or rotating at purge speed or greater and the diverter damper and Guillotine are open to allow exhaust gas flow through the HRSG. When exhaust flow is present in the HRSG, the GT status is set to an integer value of one. On shut down where the gas turbine speed is under 10%, or any condition where the exhaust flow is not through the HRSG, a two hour delay will follow before the GT status is set to 0. After two hours, the temperature difference between inner and middle temperatures will have reached equilibrium. When the online calculation receives a GT status of 0, the calculations will halt for all components of the affected HRSG. The calculation engine will close all open cycles associated with its calculations to ensure that no matter how long the HRSG is down, the online calculation engine will start new calculations on a HRSG restart.

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HP Drum	
BxHx_S1_SUSPEND	
Bad Quality	*Integer Value
TmAW_Suspend	**4**
TmBw_Suspend	**6**
TiAW_Suspend	**3**
TiBW_Suspend	**5**
P_Suspend	**7**
Shells	
BxHx_Sx_SUSPEND	
Bad Quality	*Integer Value
Tm_Suspend	**2**
Ti_Suspend	**1**
P_Suspend	**7**

*** Listed in order of priority**

TiAW - Temperature Inner Above Water
TmAW -Temperature Middle Above Water
TiBW - Temperature Inner Below Water
TmBW -Temperature Middle Below Water
Ti - Temperature Inner
Tm -Temperature Middle
P - Pressure

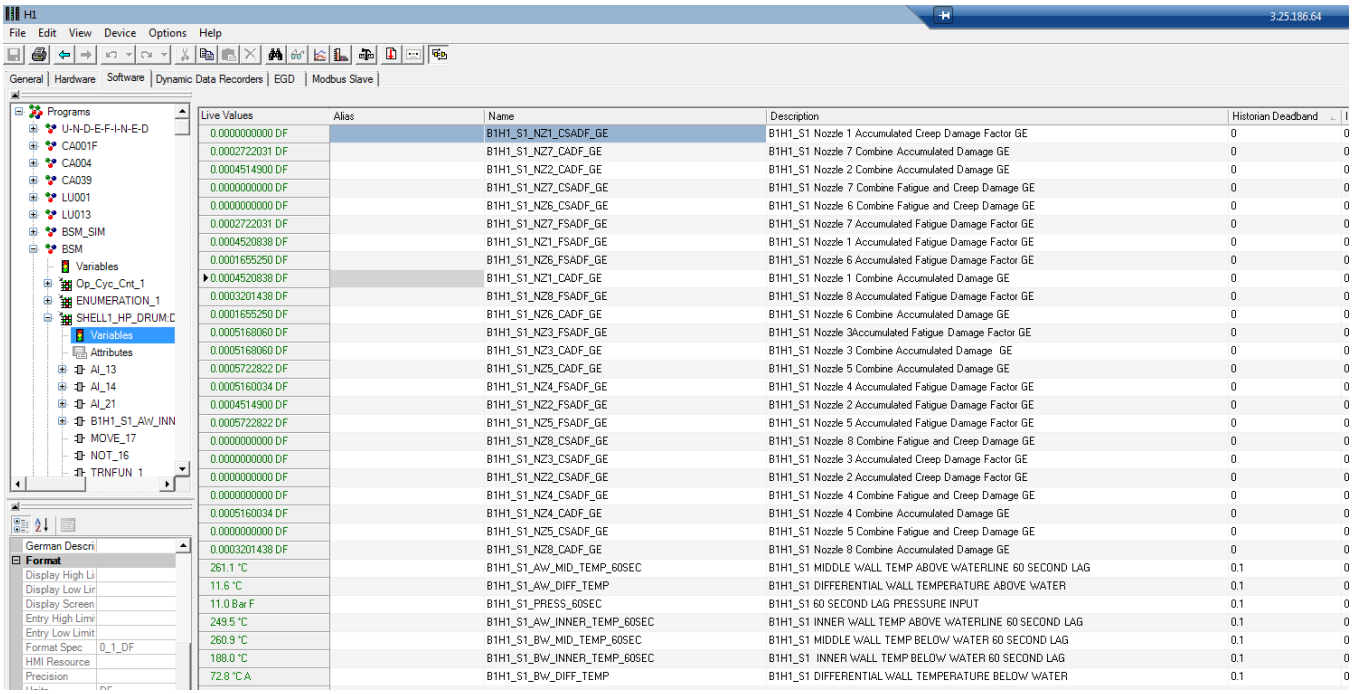


BxHx_Sx_SUSPEND Variable Possible Values

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2.3 Historian Communication

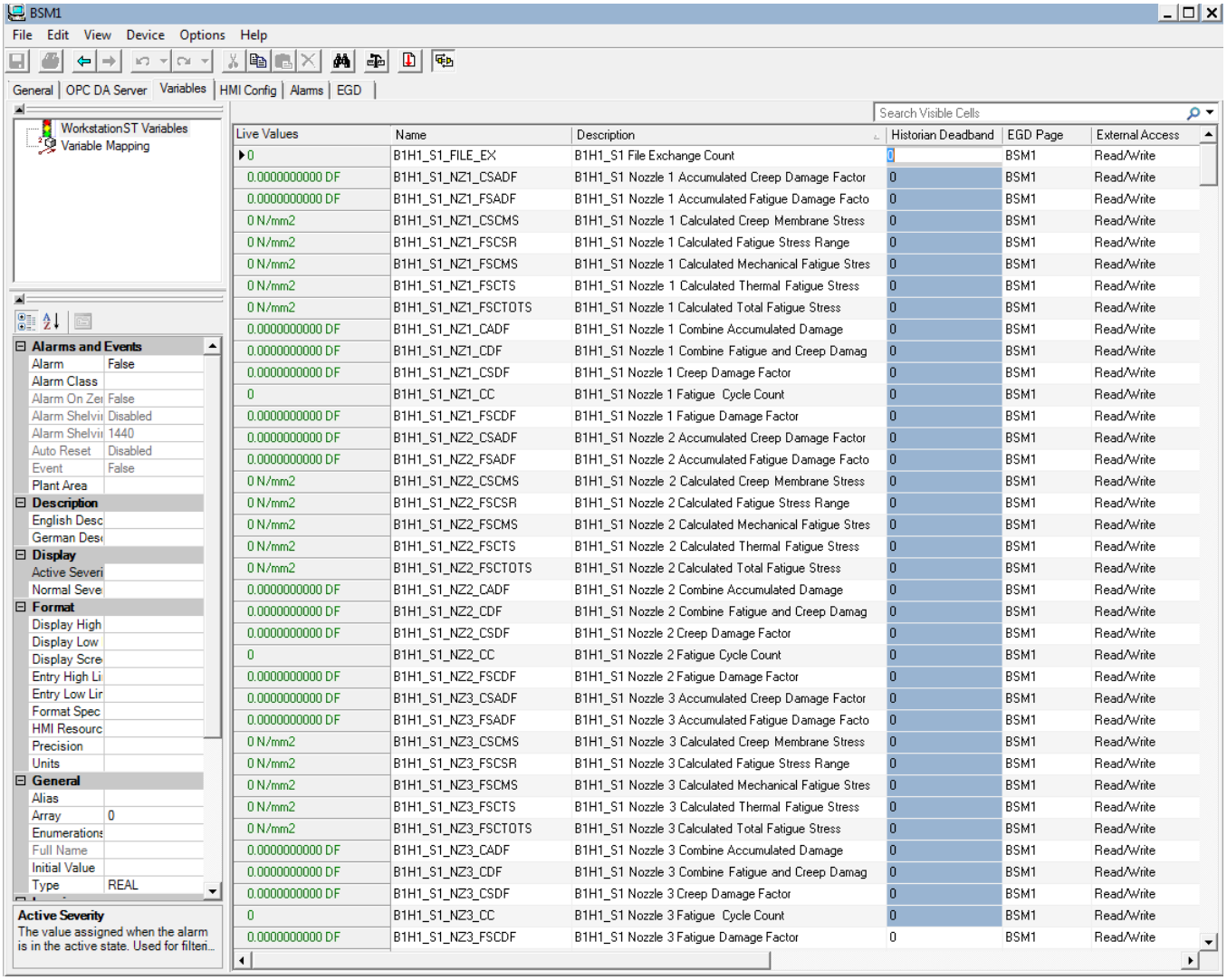
All data sent to the plant data historian originates from each local HRSG controller and the BSM1 workstation. Historian tag attributes are adjusted to indicate a historical point and placed on the EGD. The historian collects the data from the referenced device EGD pages, through the OPC DA Server with which the OSI OPC interface communicates, to pass data to the historian.



Live Values	Alias	Name	Description	Historian Deadband
0.0000000000 DF		B1H1_S1_NZ1_CSADF_GE	B1H1_S1 Nozzle 1 Accumulated Creep Damage Factor GE	0
0.0002722031 DF		B1H1_S1_NZ7_CADF_GE	B1H1_S1 Nozzle 7 Combine Accumulated Damage GE	0
0.0004514900 DF		B1H1_S1_NZ2_CADF_GE	B1H1_S1 Nozzle 2 Combine Accumulated Damage GE	0
0.0000000000 DF		B1H1_S1_NZ7_CSADF_GE	B1H1_S1 Nozzle 7 Combine Fatigue and Creep Damage GE	0
0.0000000000 DF		B1H1_S1_NZ6_CSADF_GE	B1H1_S1 Nozzle 6 Combine Fatigue and Creep Damage GE	0
0.0002722031 DF		B1H1_S1_NZ7_FSADF_GE	B1H1_S1 Nozzle 7 Accumulated Fatigue Damage Factor GE	0
0.0004520838 DF		B1H1_S1_NZ1_FSADF_GE	B1H1_S1 Nozzle 1 Accumulated Fatigue Damage Factor GE	0
0.0001655250 DF		B1H1_S1_NZ6_FSADF_GE	B1H1_S1 Nozzle 6 Accumulated Fatigue Damage Factor GE	0
0.0004520838 DF		B1H1_S1_NZ1_CADF_GE	B1H1_S1 Nozzle 1 Combine Accumulated Damage GE	0
0.0003201438 DF		B1H1_S1_NZ8_FSADF_GE	B1H1_S1 Nozzle 8 Accumulated Fatigue Damage Factor GE	0
0.0001655250 DF		B1H1_S1_NZ6_CADF_GE	B1H1_S1 Nozzle 6 Combine Accumulated Damage GE	0
0.0005168060 DF		B1H1_S1_NZ3_FSADF_GE	B1H1_S1 Nozzle 3Accumulated Fatigue Damage Factor GE	0
0.0005168060 DF		B1H1_S1_NZ3_CADF_GE	B1H1_S1 Nozzle 3 Combine Accumulated Damage GE	0
0.0005722822 DF		B1H1_S1_NZ5_CADF_GE	B1H1_S1 Nozzle 5 Combine Accumulated Damage GE	0
0.0005160034 DF		B1H1_S1_NZ4_FSADF_GE	B1H1_S1 Nozzle 4 Accumulated Fatigue Damage Factor GE	0
0.0004514900 DF		B1H1_S1_NZ2_FSADF_GE	B1H1_S1 Nozzle 2 Accumulated Fatigue Damage Factor GE	0
0.0005722822 DF		B1H1_S1_NZ5_FSADF_GE	B1H1_S1 Nozzle 5 Accumulated Fatigue Damage Factor GE	0
0.0000000000 DF		B1H1_S1_NZ8_CSADF_GE	B1H1_S1 Nozzle 8 Combine Fatigue and Creep Damage GE	0
0.0000000000 DF		B1H1_S1_NZ3_CSADF_GE	B1H1_S1 Nozzle 3 Accumulated Creep Damage Factor GE	0
0.0000000000 DF		B1H1_S1_NZ2_CSADF_GE	B1H1_S1 Nozzle 2 Accumulated Creep Damage Factor GE	0
0.0000000000 DF		B1H1_S1_NZ4_CSADF_GE	B1H1_S1 Nozzle 4 Combine Fatigue and Creep Damage GE	0
0.0005160034 DF		B1H1_S1_NZ4_CADF_GE	B1H1_S1 Nozzle 4 Combine Accumulated Damage GE	0
0.0000000000 DF		B1H1_S1_NZ5_CSADF_GE	B1H1_S1 Nozzle 5 Combine Fatigue and Creep Damage GE	0
0.0003201438 DF		B1H1_S1_NZ8_CADF_GE	B1H1_S1 Nozzle 8 Combine Accumulated Damage GE	0
261.1 °C		B1H1_S1_AW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1
11.6 °C		B1H1_S1_AW_DIFF_TEMP	B1H1_S1 DIFFERENTIAL WALL TEMPERATURE ABOVE WATER	0.1
11.0 Bar F		B1H1_S1_PRESS_60SEC	B1H1_S1 60 SECOND LAG PRESSURE INPUT	0.1
249.5 °C		B1H1_S1_AW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1
260.9 °C		B1H1_S1_BW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP BELOW WATER 60 SECOND LAG	0.1
188.0 °C		B1H1_S1_BW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP BELOW WATER 60 SECOND LAG	0.1
72.8 °C A		B1H1_S1_BW_DIFF_TEMP	B1H1_S1 DIFFERENTIAL WALL TEMPERATURE BELOW WATER	0.1

Typical Controller-based Variables Assigned as Historical Points

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Example of BSM1 Workstation Client Driven Variables Configured for Historian

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3 DXT Calculation Engine

Creep damage is determined using material creep properties, as well as local stress and exposure time.

The DXT calculation engine calculates parameters for evaluating and determining instantaneous and accumulated fatigue and creep damage factors for the selected components of the HRSG system.

Based on pressure and temperature measurements coming from the DCS, local stresses and stress ranges are calculated and returned to the DCS historian for storage. From the calculated stress and stress range variation, fatigue damage can be determined using the Rain Flow/Range Pair method for cycle counting as described in EN12952-4.

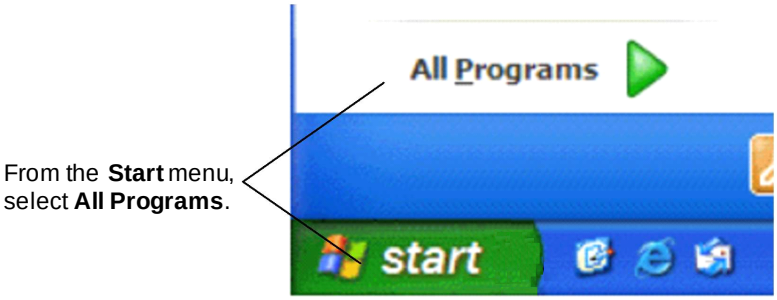
The application is pre-installed on a workstation solely dedicated to the calculation of stress and lifecycle monitoring. Normally, it automatically starts when powering on the workstation. The application can be manually started from the Windows® Start menu.

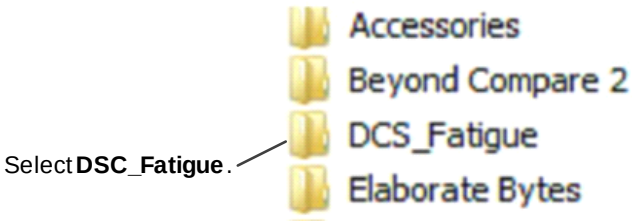


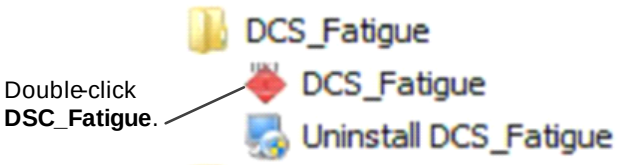
Attention

Before running the program, the HRSG component geometry, material properties, weld type, weld quality, and so forth must be specified. The component configuration is based on detailed drawings and design information from the HRSG vendor and will be made by GE prior to installing the software on site.

➢ **To start the BSM online software**







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3.1 Access Modes

The BSM program runs in three different access modes.

Access Mode	Password Required
Full access	Yes
Partial access	Yes
Monitor access	No

3.1.1 Full Access

Full access mode is restricted to DXT and GE. All program functionality is available to the user for setting up new projects, including:

- Input for component geometry, material, mechanical strength, stress concentration factors, and so forth
- Tag name definition and data exchange setup for communication with the DCS

Expertise is required for configuration. Only a user with full access to the system will be able to view the configuration screens.



Caution

Any modifications can affect the algorithms of the calculation engine, rendering the output of the online software incorrect. Once the proper configuration for the BSM is completed by GE, it must not be altered.

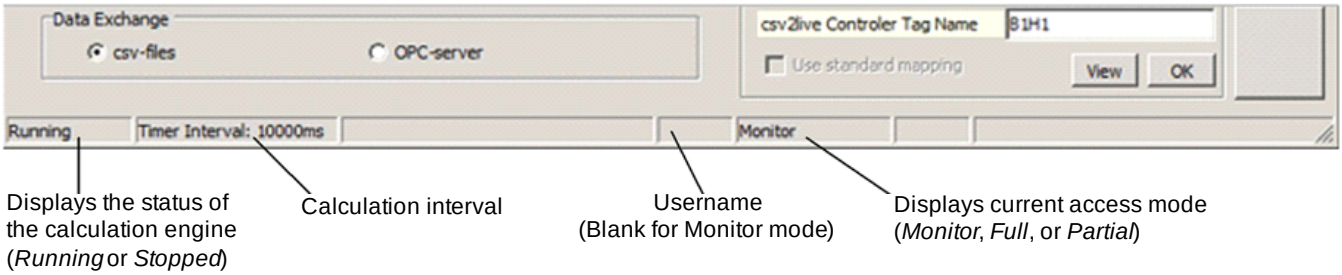
3.1.2 Partial Access

Partial software access is restricted to GE and gives permission for:

- Tag name definition and data exchange setup for communication with the DCS
- Timer value for the frequency of calculations

3.1.3 Monitor Access

Monitor access is the default mode. It allows the user to view individual menus but not change any settings. The status bar in the bottom of the main screen displays the status of the online software.



Monitor Access Main Screen

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3.2 Start Up

The DXT calculation engine runs in Monitor mode at start up. To change any settings or the HRSG configuration, the user must log on with a username and password provided by GE. The username and password determine the access level permissions available for the user.

Note GE restricts access of the software configuration parameters to GE personnel.

During start up, the configuration data specific to the actual site is automatically loaded and starts the communication with the DCS. Data exchange between the HRSG Stress and LifeCycle Monitor and the DCS can be performed in two ways:

Both communication methods require unique configuration and are not interchangeable on site. Refer to the section, Live2csv.

- Using comma separated files (.csv files).
- Using the WorkstationST OPC DA Server. The online software (DXT) acts as an OPC client.

Key features include:

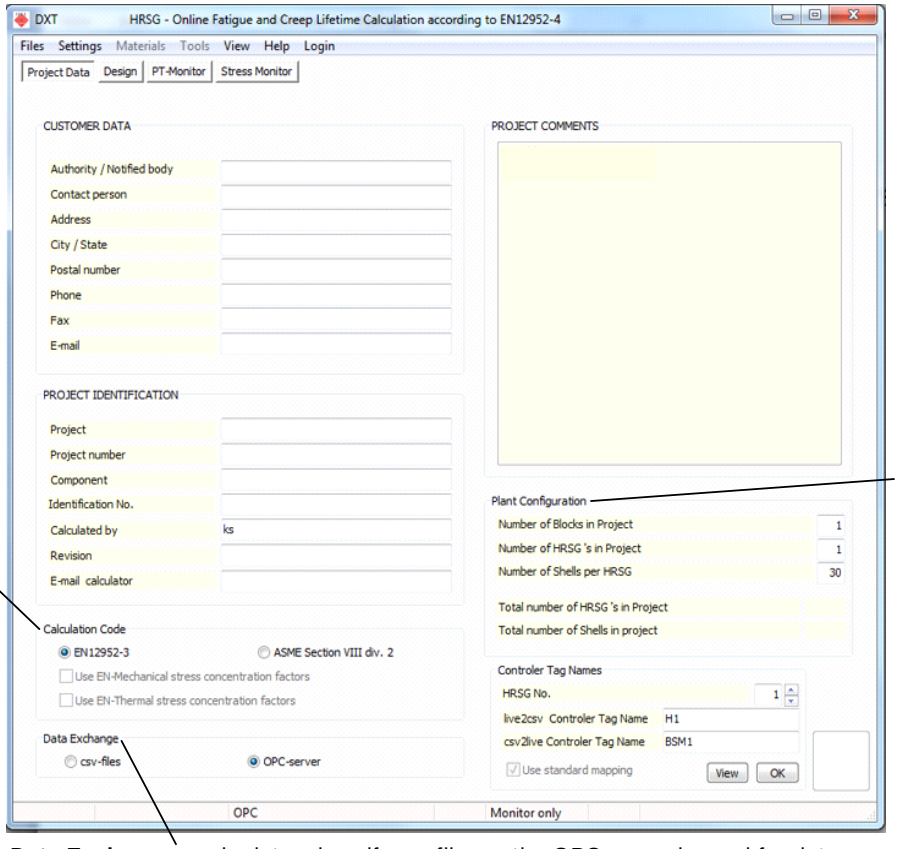
- The actual data exchange is determined by GE before installing the software.
- If the workstation is accidentally shut down, the software starts up and runs in Monitor mode on reboot.
- If the communication between the online software and the DCS is lost, the software will keep trying to connect to the DCS provided .csv files or OPC server.
- The communication status of *Running* or *Stopped* is displayed in the status bar.

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3.3 Software Screens

3.3.1 Project Data

The Project Data screen contains information on the actual project setup. Modification of specific parameters are password controlled.



Calculation Code determines if the stress calculation is based on the EN12952-3 standard or the ASME Section VIII div. 2 standard. The EN-12952-2 standard is the default selection.

Data Exchange mode determines if .csv files or the OPC server is used for data exchange between the DCS and BSM software.

Plant Configuration is used for Tag Name mapping to define the data exchange between the DCS and calculation parameters.

Based on the actual numbers of blocks, HRSGs, and shells, the necessary Tag Names will automatically generate using controller tag names. They are listed when the user is logged on with either *Partial* or *Full* access by pressing **View**.

3.3.2 PT-Monitor

The PT-Monitor screen displays a graphical representation of the last 100 measurements of inner and middle wall temperatures and water/steam pressure for each individual monitored component shell. The actual shell can be selected in the menu located below the chart.

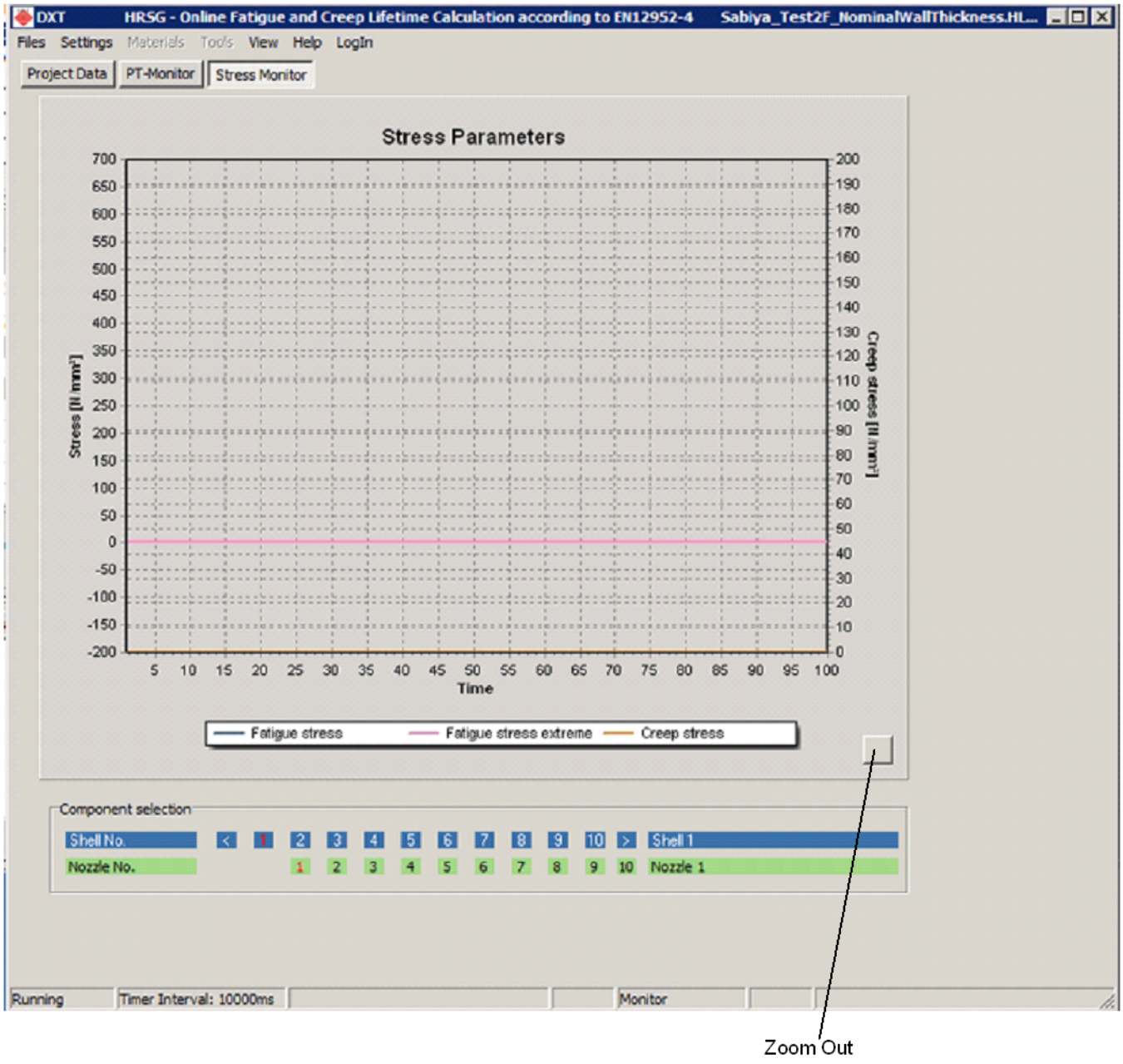
Tip *Zoom in by clicking the left-mouse button while drawing a rectangle on the graph.*

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3.3.3 Stress Monitor

The Stress Monitor screen displays the last 100 values for calculated fatigue stress, fatigue stress extreme values, and creep stress for each individual nozzle. The shell or nozzle can be selected in the Component Selection menu.

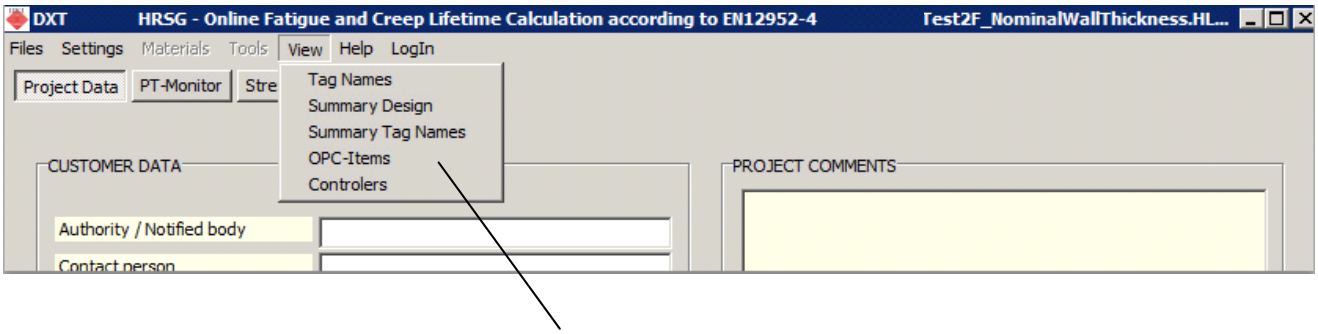
Tip *Zoom in by clicking the left-mouse button while drawing a rectangle on the graph.*



Stress Monitor Screen

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3.3.4 View Menu



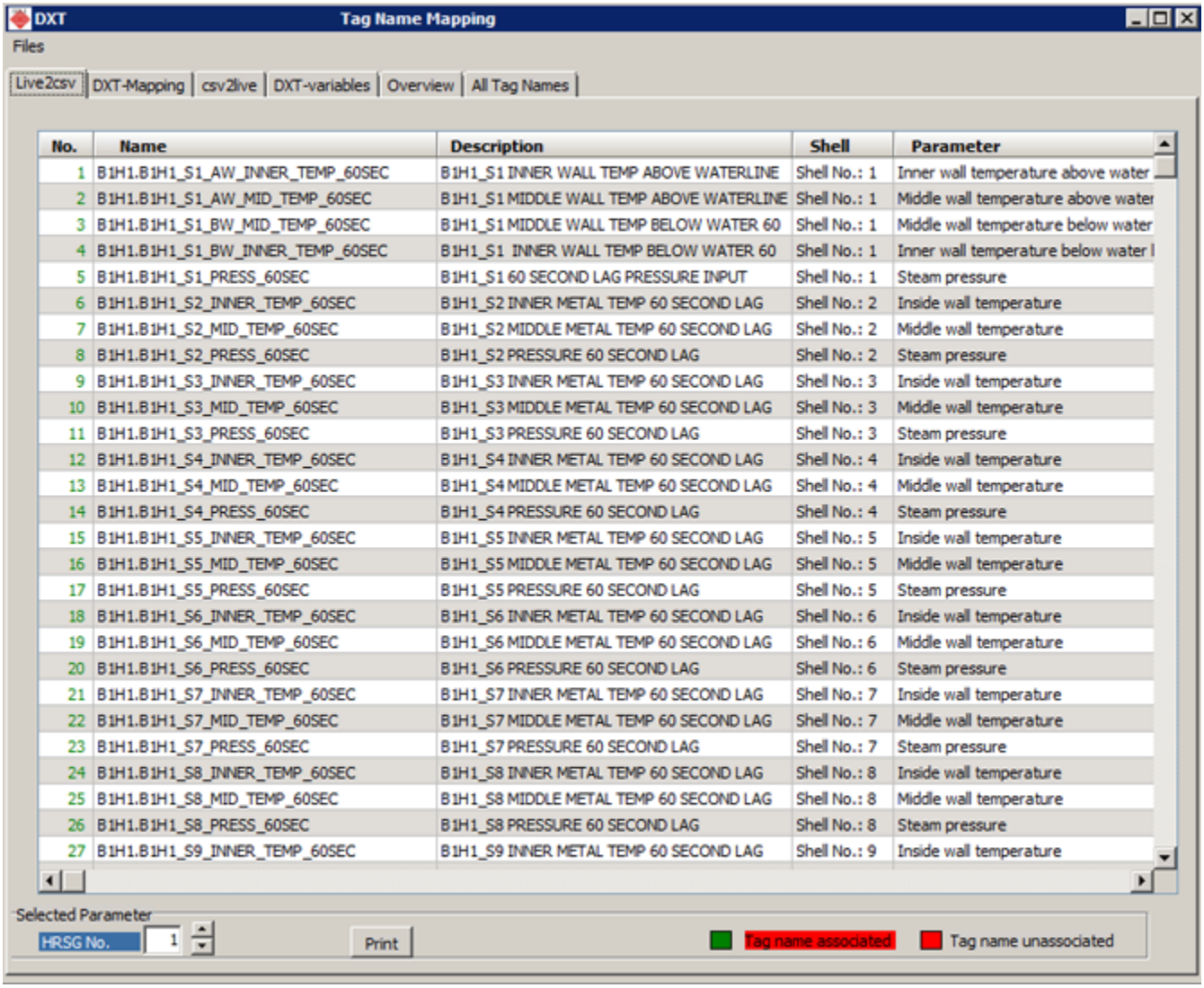
The **View** menu provides access to different design and online parts.

3.4 Tag Name Mapping Screens

3.4.1 Live2csv

The Live2csv screen displays the status for the Tag Name mapping of parameters and values coming from the DCS to the BSM software. A green number in the first column indicates that the actual Tag Name has an association. A red number indicates that no association is found for that specific Tag Name. If the project contains more than one HRSG, different HRSGs can be selected in the Selected Parameter menu. Columns can be scaled by dragging the headings.

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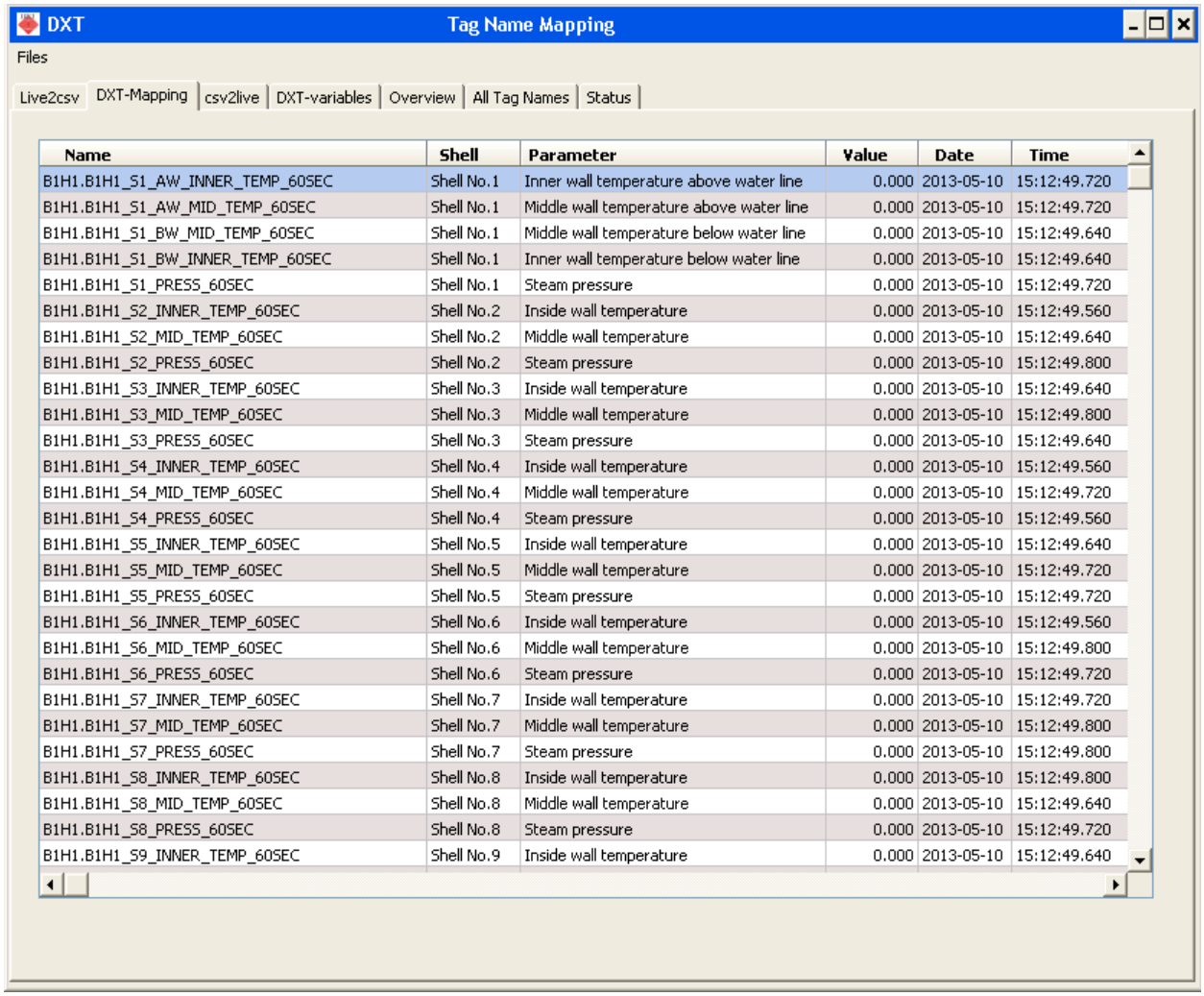
No.	Name	Description	Shell	Parameter
1	B1H1.B1H1_S1_AW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP ABOVE WATERLINE	Shell No.: 1	Inner wall temperature above water
2	B1H1.B1H1_S1_AW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP ABOVE WATERLINE	Shell No.: 1	Middle wall temperature above wate
3	B1H1.B1H1_S1_BW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP BELOW WATER 60	Shell No.: 1	Middle wall temperature below water
4	B1H1.B1H1_S1_BW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP BELOW WATER 60	Shell No.: 1	Inner wall temperature below water l
5	B1H1.B1H1_S1_PRESS_60SEC	B1H1_S1 60 SECOND LAG PRESSURE INPUT	Shell No.: 1	Steam pressure
6	B1H1.B1H1_S2_INNER_TEMP_60SEC	B1H1_S2 INNER METAL TEMP 60 SECOND LAG	Shell No.: 2	Inside wall temperature
7	B1H1.B1H1_S2_MID_TEMP_60SEC	B1H1_S2 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 2	Middle wall temperature
8	B1H1.B1H1_S2_PRESS_60SEC	B1H1_S2 PRESSURE 60 SECOND LAG	Shell No.: 2	Steam pressure
9	B1H1.B1H1_S3_INNER_TEMP_60SEC	B1H1_S3 INNER METAL TEMP 60 SECOND LAG	Shell No.: 3	Inside wall temperature
10	B1H1.B1H1_S3_MID_TEMP_60SEC	B1H1_S3 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 3	Middle wall temperature
11	B1H1.B1H1_S3_PRESS_60SEC	B1H1_S3 PRESSURE 60 SECOND LAG	Shell No.: 3	Steam pressure
12	B1H1.B1H1_S4_INNER_TEMP_60SEC	B1H1_S4 INNER METAL TEMP 60 SECOND LAG	Shell No.: 4	Inside wall temperature
13	B1H1.B1H1_S4_MID_TEMP_60SEC	B1H1_S4 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 4	Middle wall temperature
14	B1H1.B1H1_S4_PRESS_60SEC	B1H1_S4 PRESSURE 60 SECOND LAG	Shell No.: 4	Steam pressure
15	B1H1.B1H1_S5_INNER_TEMP_60SEC	B1H1_S5 INNER METAL TEMP 60 SECOND LAG	Shell No.: 5	Inside wall temperature
16	B1H1.B1H1_S5_MID_TEMP_60SEC	B1H1_S5 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 5	Middle wall temperature
17	B1H1.B1H1_S5_PRESS_60SEC	B1H1_S5 PRESSURE 60 SECOND LAG	Shell No.: 5	Steam pressure
18	B1H1.B1H1_S6_INNER_TEMP_60SEC	B1H1_S6 INNER METAL TEMP 60 SECOND LAG	Shell No.: 6	Inside wall temperature
19	B1H1.B1H1_S6_MID_TEMP_60SEC	B1H1_S6 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 6	Middle wall temperature
20	B1H1.B1H1_S6_PRESS_60SEC	B1H1_S6 PRESSURE 60 SECOND LAG	Shell No.: 6	Steam pressure
21	B1H1.B1H1_S7_INNER_TEMP_60SEC	B1H1_S7 INNER METAL TEMP 60 SECOND LAG	Shell No.: 7	Inside wall temperature
22	B1H1.B1H1_S7_MID_TEMP_60SEC	B1H1_S7 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 7	Middle wall temperature
23	B1H1.B1H1_S7_PRESS_60SEC	B1H1_S7 PRESSURE 60 SECOND LAG	Shell No.: 7	Steam pressure
24	B1H1.B1H1_S8_INNER_TEMP_60SEC	B1H1_S8 INNER METAL TEMP 60 SECOND LAG	Shell No.: 8	Inside wall temperature
25	B1H1.B1H1_S8_MID_TEMP_60SEC	B1H1_S8 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 8	Middle wall temperature
26	B1H1.B1H1_S8_PRESS_60SEC	B1H1_S8 PRESSURE 60 SECOND LAG	Shell No.: 8	Steam pressure
27	B1H1.B1H1_S9_INNER_TEMP_60SEC	B1H1_S9 INNER METAL TEMP 60 SECOND LAG	Shell No.: 9	Inside wall temperature

Live2csv Screen

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3.4.2 DXT Mapping

The DXT Mapping screen displays Tag Name Mapping and parameter association for variables coming from the DCS to the BSM software.



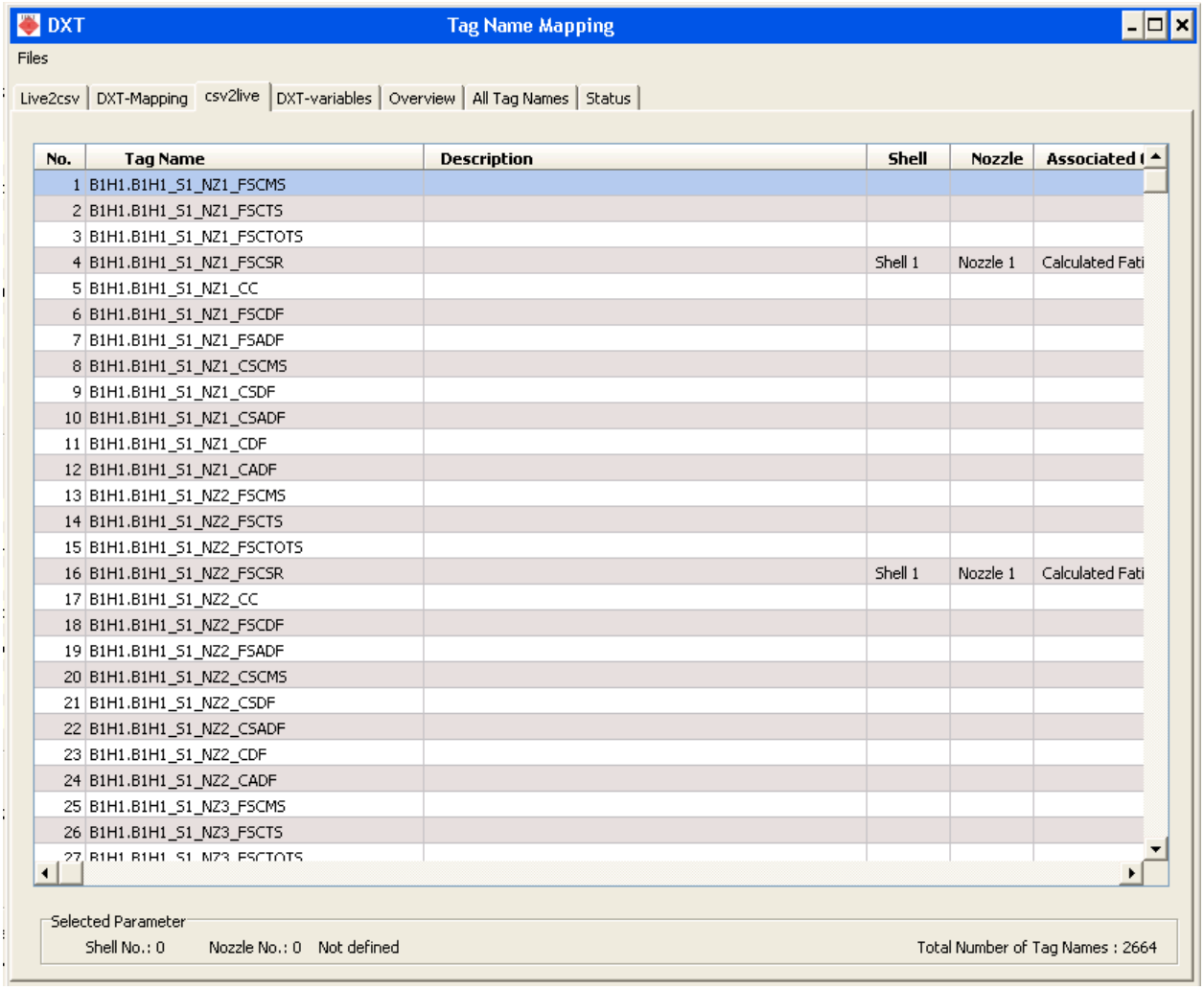
Name	Shell	Parameter	Value	Date	Time
B1H1.B1H1_S1_AW_INNER_TEMP_60SEC	Shell No.1	Inner wall temperature above water line	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S1_AW_MID_TEMP_60SEC	Shell No.1	Middle wall temperature above water line	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S1_BW_MID_TEMP_60SEC	Shell No.1	Middle wall temperature below water line	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S1_BW_INNER_TEMP_60SEC	Shell No.1	Inner wall temperature below water line	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S1_PRESS_60SEC	Shell No.1	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S2_INNER_TEMP_60SEC	Shell No.2	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S2_MID_TEMP_60SEC	Shell No.2	Middle wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S2_PRESS_60SEC	Shell No.2	Steam pressure	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S3_INNER_TEMP_60SEC	Shell No.3	Inside wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S3_MID_TEMP_60SEC	Shell No.3	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S3_PRESS_60SEC	Shell No.3	Steam pressure	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S4_INNER_TEMP_60SEC	Shell No.4	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S4_MID_TEMP_60SEC	Shell No.4	Middle wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S4_PRESS_60SEC	Shell No.4	Steam pressure	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S5_INNER_TEMP_60SEC	Shell No.5	Inside wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S5_MID_TEMP_60SEC	Shell No.5	Middle wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S5_PRESS_60SEC	Shell No.5	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S6_INNER_TEMP_60SEC	Shell No.6	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S6_MID_TEMP_60SEC	Shell No.6	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S6_PRESS_60SEC	Shell No.6	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S7_INNER_TEMP_60SEC	Shell No.7	Inside wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S7_MID_TEMP_60SEC	Shell No.7	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S7_PRESS_60SEC	Shell No.7	Steam pressure	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S8_INNER_TEMP_60SEC	Shell No.8	Inside wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S8_MID_TEMP_60SEC	Shell No.8	Middle wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S8_PRESS_60SEC	Shell No.8	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S9_INNER_TEMP_60SEC	Shell No.9	Inside wall temperature	0.000	2013-05-10	15:12:49.640

DXT Mapping Screen

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3.4.3 Csv2live

The Csv2live screen displays the status for the Tag Name mapping of parameters and values going from the BSM software to the DCS.

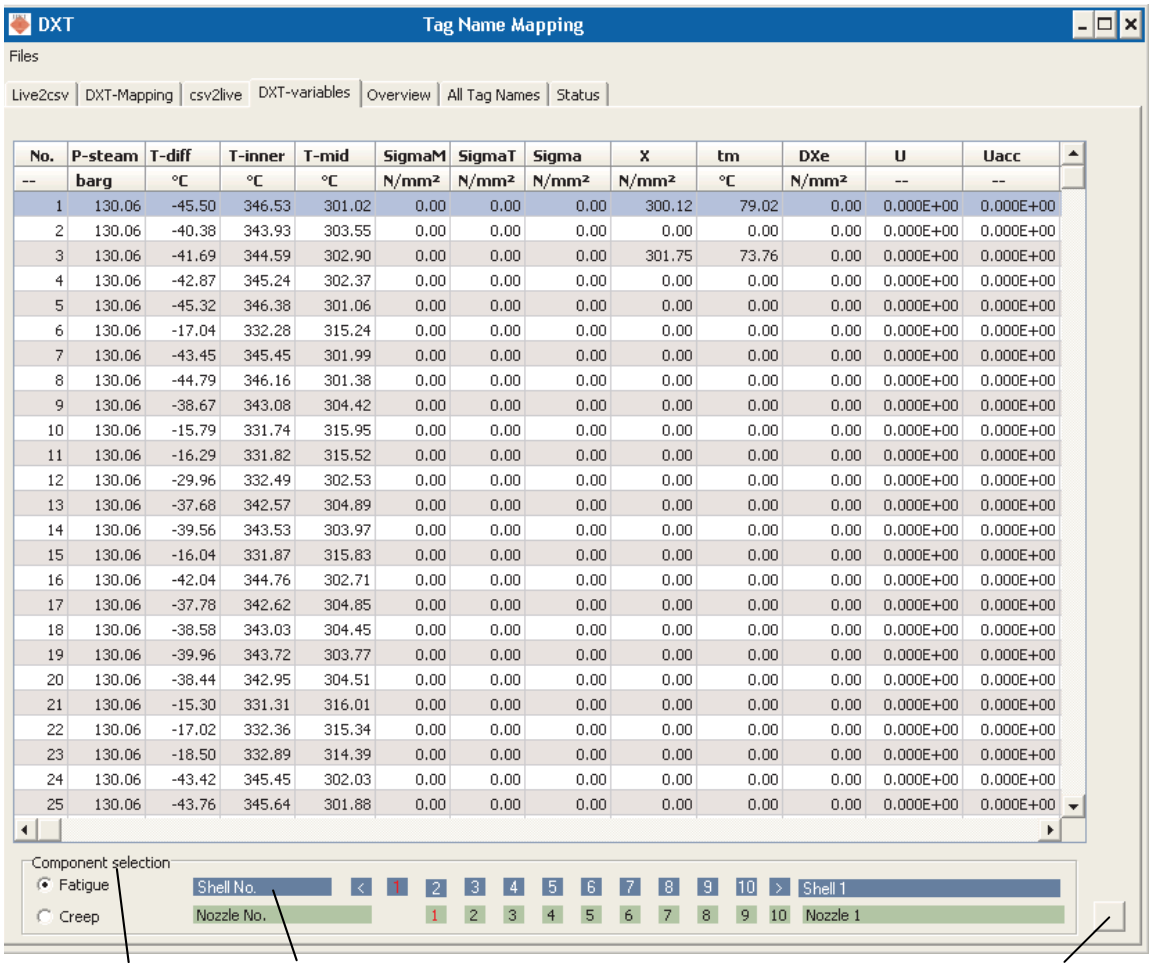


Csv2live Screen

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3.4.4 DXT–Variables

The DXT-Variables screen displays parameters used for fatigue and creep damage calculation for all monitored nozzle shells. The grid contains values for the latest 100 calculations and is updated every time the data exchange between the DCS and BSM software takes place. Individual nozzles on the monitored components can be selected in the Component Selection menu. The **Fatigue** and **Creep** buttons display or hide fatigue and creep parameters. Placing the cursor at the column headers provides a more detailed explanation and engineering units of the parameters listed in the grid.



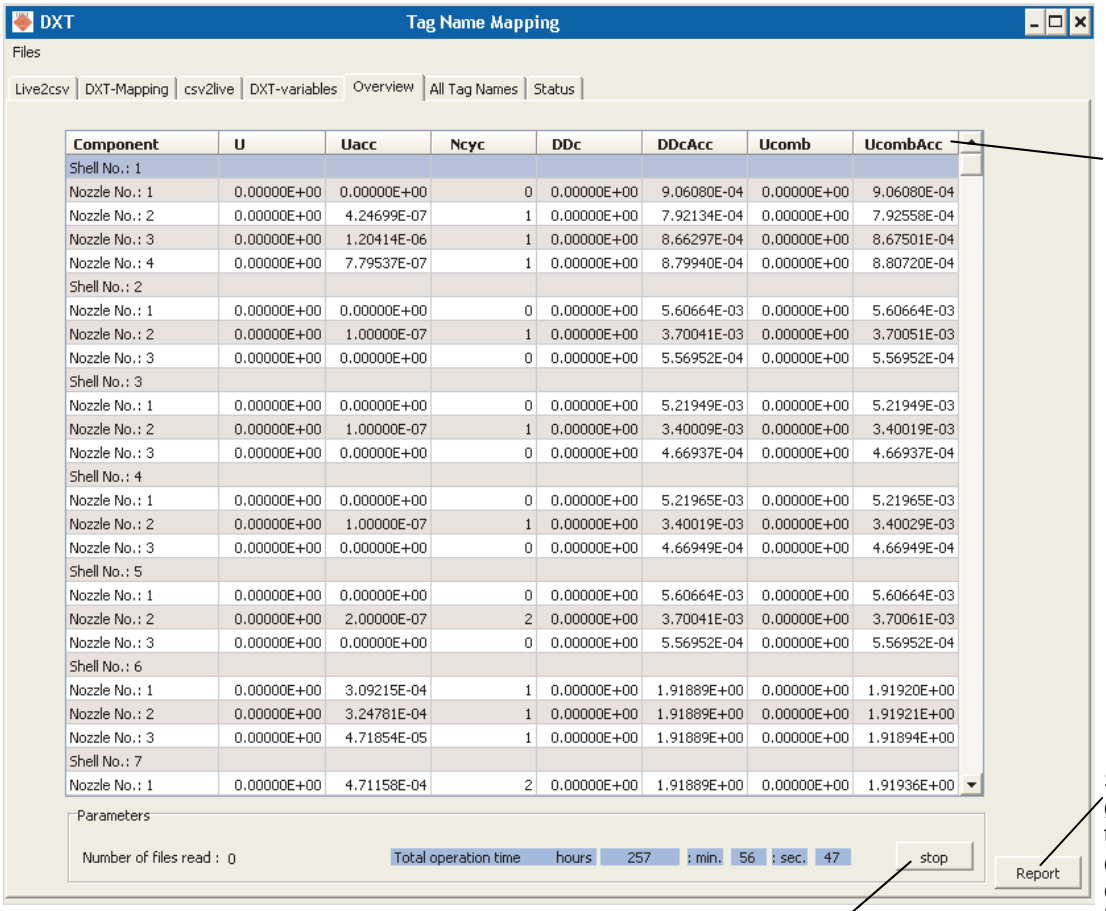
No.	P-steam	T-diff	T-inner	T-mid	SigmaM	SigmaT	Sigma	X	tm	DXe	U	Uacc
--	barg	°C	°C	°C	N/mm²	N/mm²	N/mm²	N/mm²	°C	N/mm²	--	--
1	130.06	-45.50	346.53	301.02	0.00	0.00	0.00	300.12	79.02	0.00	0.000E+00	0.000E+00
2	130.06	-40.38	343.93	303.55	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
3	130.06	-41.69	344.59	302.90	0.00	0.00	0.00	301.75	73.76	0.00	0.000E+00	0.000E+00
4	130.06	-42.87	345.24	302.37	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
5	130.06	-45.32	346.38	301.06	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
6	130.06	-17.04	332.28	315.24	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
7	130.06	-43.45	345.45	301.99	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
8	130.06	-44.79	346.16	301.38	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
9	130.06	-38.67	343.08	304.42	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
10	130.06	-15.79	331.74	315.95	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
11	130.06	-16.29	331.82	315.52	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
12	130.06	-29.96	332.49	302.53	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
13	130.06	-37.68	342.57	304.89	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
14	130.06	-39.56	343.53	303.97	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
15	130.06	-16.04	331.87	315.83	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
16	130.06	-42.04	344.76	302.71	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
17	130.06	-37.78	342.62	304.85	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
18	130.06	-38.58	343.03	304.45	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
19	130.06	-39.96	343.72	303.77	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
20	130.06	-38.44	342.95	304.51	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
21	130.06	-15.30	331.31	316.01	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
22	130.06	-17.02	332.36	315.34	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
23	130.06	-18.50	332.89	314.39	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
24	130.06	-43.42	345.45	302.03	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
25	130.06	-43.76	345.64	301.88	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00

DXT-Variables Screen

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3.4.5 Overview Tab

The Overview tab displays the calculated values for instant and accumulated fatigue and creep damage factions. Reports can be generated containing the fatigue and creep damage factions being monitored. The grid displays the latest 100 calculations. Fatigue damage is based on the last 100 cycles and the creep damage is accumulated from the last time the software was started to the present. Accumulated values are stored in the plant data historian.



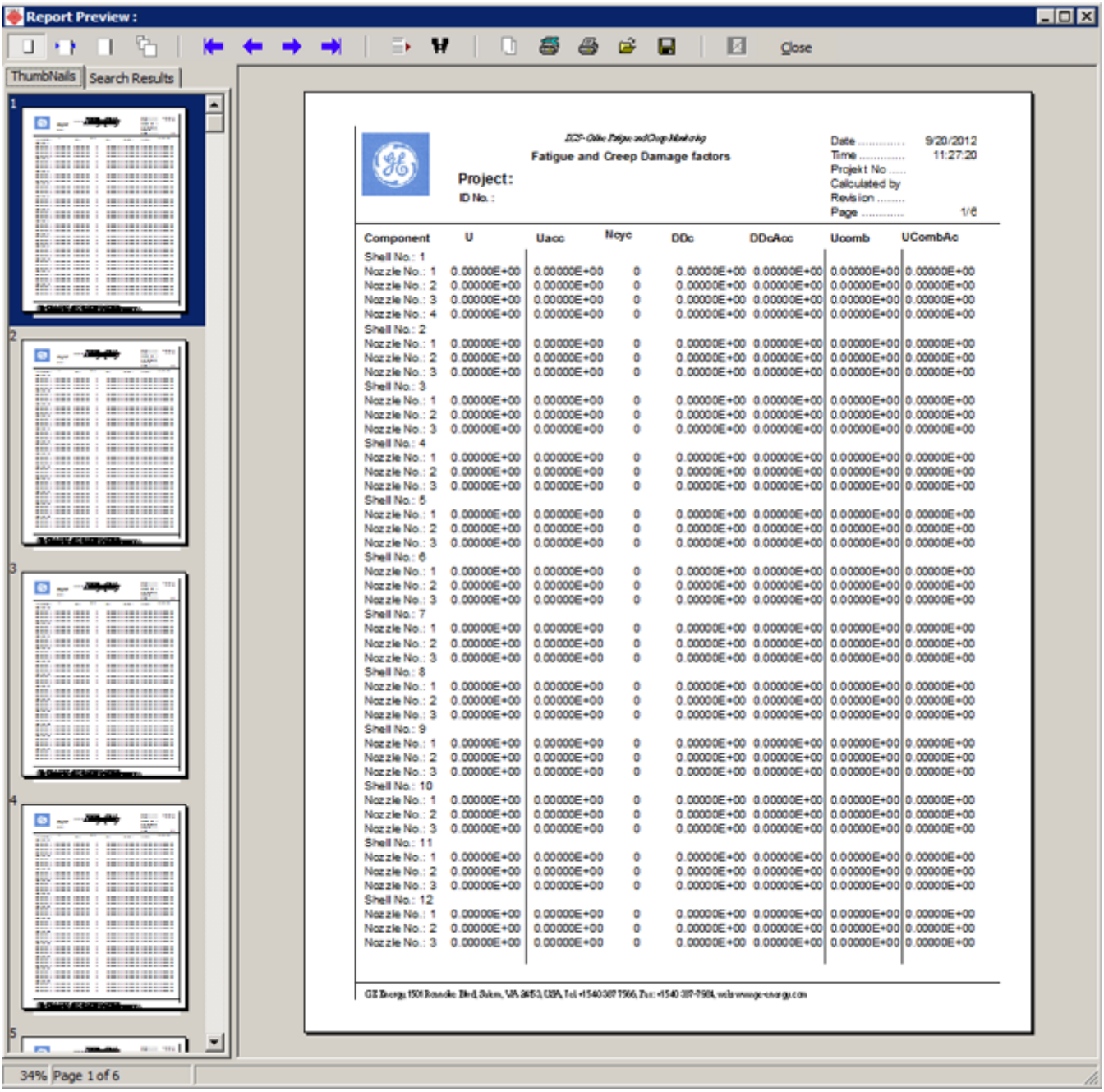
Placing the cursor at the column header provides a more detailed explanation and engineering units of the parameters listed in the grid.

Select **Report** to generate a report of the fatigue and creep damage fraction for the component nozzles being monitored.

Select **Stop** to suspend the parameter update.

Overview Screen

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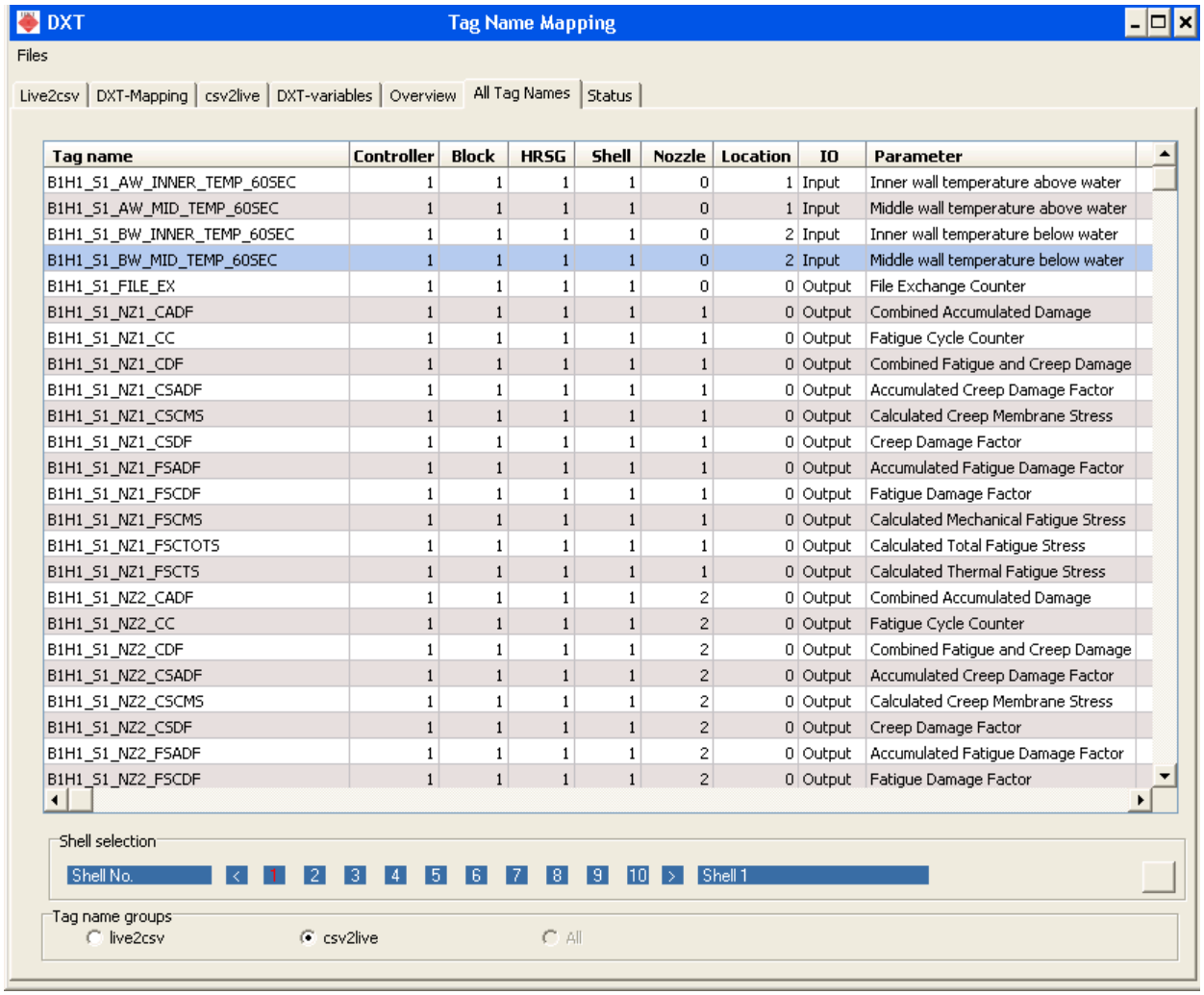


Example of Report Preview

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3.4.6 All Tag Names Tab

The All Tag Names tab displays a list of all parameters and variables with attributes consumed and produced by the software, site block, HRSG, shells, and nozzles. The attributes identify the component and sub-component, if applicable, with which the parameter or variable is associated, whether it is an input (live2csv) or output (csv2live), and a description of the parameter or variable. Inputs and outputs from the BSM can be selected from the Tag name group menu.



Tag name	Controller	Block	HRSG	Shell	Nozzle	Location	IO	Parameter
B1H1_S1_AW_INNER_TEMP_60SEC	1	1	1	1	0	1	Input	Inner wall temperature above water
B1H1_S1_AW_MID_TEMP_60SEC	1	1	1	1	0	1	Input	Middle wall temperature above water
B1H1_S1_BW_INNER_TEMP_60SEC	1	1	1	1	0	2	Input	Inner wall temperature below water
B1H1_S1_BW_MID_TEMP_60SEC	1	1	1	1	0	2	Input	Middle wall temperature below water
B1H1_S1_FILE_EX	1	1	1	1	0	0	Output	File Exchange Counter
B1H1_S1_NZ1_CADF	1	1	1	1	1	0	Output	Combined Accumulated Damage
B1H1_S1_NZ1_CC	1	1	1	1	1	0	Output	Fatigue Cycle Counter
B1H1_S1_NZ1_CDF	1	1	1	1	1	0	Output	Combined Fatigue and Creep Damage
B1H1_S1_NZ1_CSADF	1	1	1	1	1	0	Output	Accumulated Creep Damage Factor
B1H1_S1_NZ1_CSCMS	1	1	1	1	1	0	Output	Calculated Creep Membrane Stress
B1H1_S1_NZ1_CSDF	1	1	1	1	1	0	Output	Creep Damage Factor
B1H1_S1_NZ1_FSADF	1	1	1	1	1	0	Output	Accumulated Fatigue Damage Factor
B1H1_S1_NZ1_FSCDF	1	1	1	1	1	0	Output	Fatigue Damage Factor
B1H1_S1_NZ1_FSCMS	1	1	1	1	1	0	Output	Calculated Mechanical Fatigue Stress
B1H1_S1_NZ1_FSCTOTS	1	1	1	1	1	0	Output	Calculated Total Fatigue Stress
B1H1_S1_NZ1_FSCTS	1	1	1	1	1	0	Output	Calculated Thermal Fatigue Stress
B1H1_S1_NZ2_CADF	1	1	1	1	2	0	Output	Combined Accumulated Damage
B1H1_S1_NZ2_CC	1	1	1	1	2	0	Output	Fatigue Cycle Counter
B1H1_S1_NZ2_CDF	1	1	1	1	2	0	Output	Combined Fatigue and Creep Damage
B1H1_S1_NZ2_CSADF	1	1	1	1	2	0	Output	Accumulated Creep Damage Factor
B1H1_S1_NZ2_CSCMS	1	1	1	1	2	0	Output	Calculated Creep Membrane Stress
B1H1_S1_NZ2_CSDF	1	1	1	1	2	0	Output	Creep Damage Factor
B1H1_S1_NZ2_FSADF	1	1	1	1	2	0	Output	Accumulated Fatigue Damage Factor
B1H1_S1_NZ2_FSCDF	1	1	1	1	2	0	Output	Fatigue Damage Factor

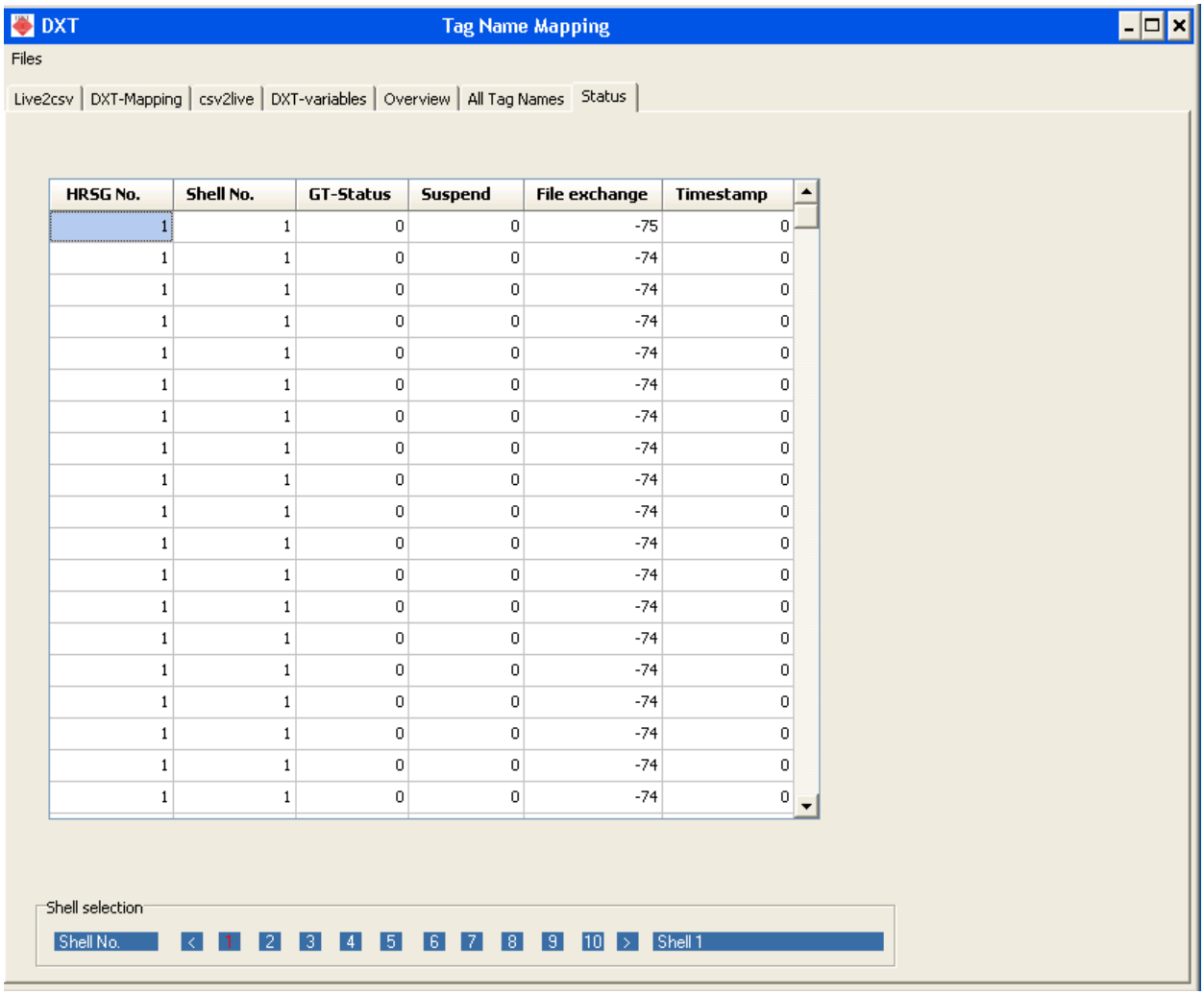
All Tag Names Screen

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3.4.7 Status Tab

Refer to the section, Data Transfer Between the Controller and DXT calculation Engine, for a matrix of the suspend parameter.

The Status tab displays a list of variables and their status as related to communications with the Mark VIe DCS. The GT-Status represents the running state of the gas turbine for the particular HRSG identified and is consumed by the online software. The Suspend variable, also consumed by the online software, is a command from the Mark VIe DCS for the halt of calculations for that particular shell of that particular HRSG. The File-exchange variable is produced by the online software and represents feedback from the online software to the Mark VIe DCS on the status of the calculations for the particular shell of a particular HRSG.

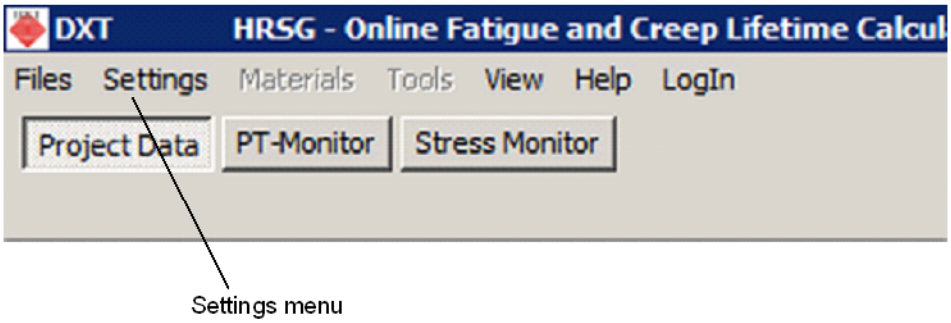


Status Screen

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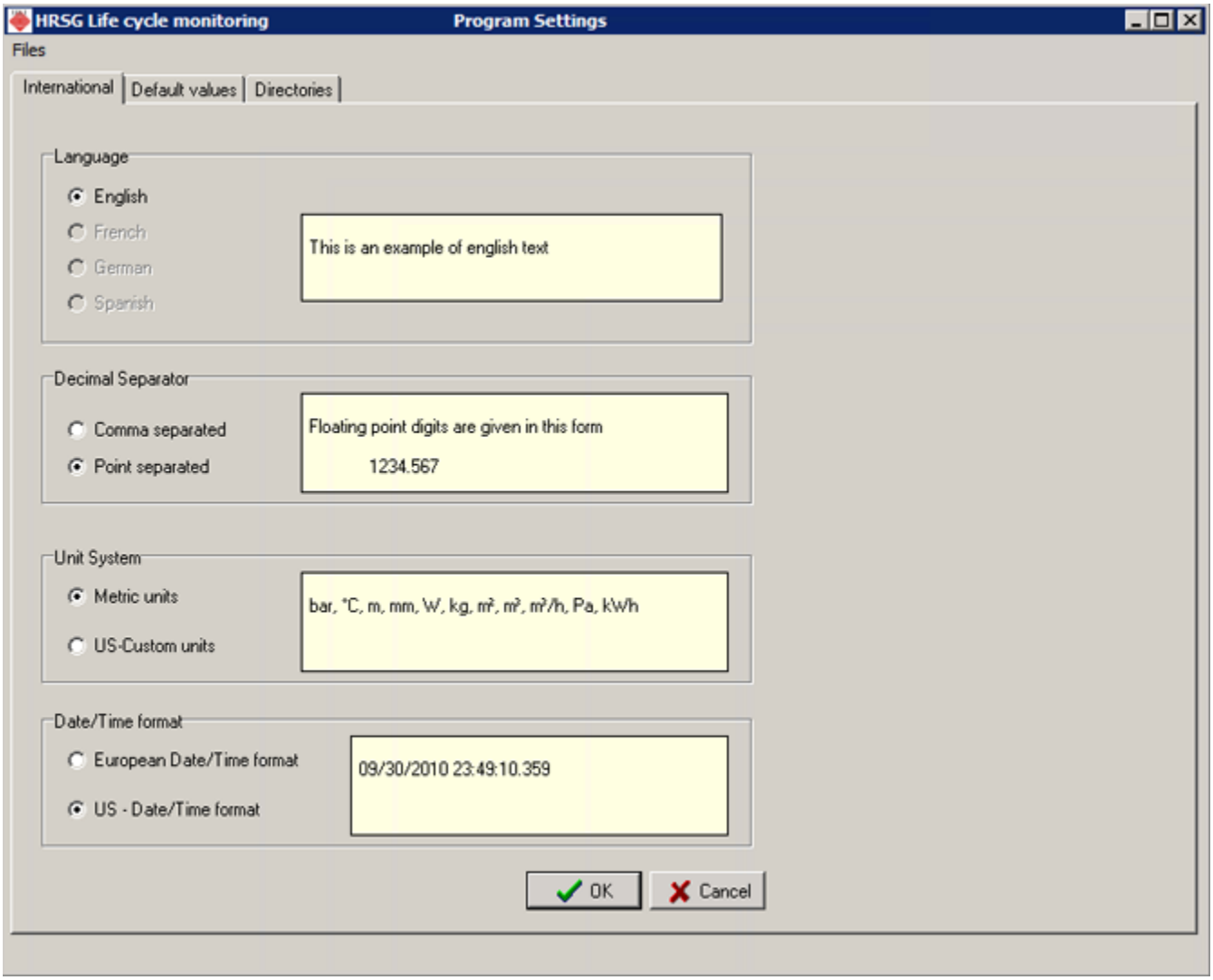
3.5 Program Settings

Program settings can be changed and stored from the Settings menu. Users can select either the International, Default values, or Directories screens.



3.5.1 International Tab

The International tab provides options for controls language display and data formatting, such as decimal separator, unit display, and time/date format.



International Screen

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The following sections are displayed on the International screen:

- **Language** is available in English only. This setting cannot be modified in Monitor mode.
- **Decimal Separator** determines formatting used when reading or writing floating point values to the screen. It is important that the decimal separator is set equal to the decimal separator in the Windows operating system (OS) to correctly read values from the screen. This setting cannot be modified in Monitor mode.
- **Unit Split** toggles between Metric and US-Custom engineering unit for all parameters in the user interface. The measurement unit can be changed by selecting the units and pressing **OK**.
- **Date/Time Format** sets the date and time format for the time stamp exchange between the DCS and BSM software when using .csv files for data exchange. It is important that the date/time format is set equal to the date/time format used in Windows OS. The date/time setting cannot be modified in the Monitor mode. When using the OPC server for data exchange, the time stamp comes directly from the DCS in ms since January 1, 1900. In this case, the actual date/time format does not matter.

  The following abbreviations are used:

  - **dd:** day in month
  - **mm:** month
  - **yyyy:** year
  - **hh:** hour (in 24–hour format)
  - **nn:** minutes
  - **ss:** seconds

Language Display Date/Time Format

Language	Date/Time Format	Example
European	dd-mm-yyyy hh:nn:ss	08-10-2012 14:22:50
US	mm/dd/yyyy hh:nn:ss	10/08/2012 14:22:50

➢ To apply settings

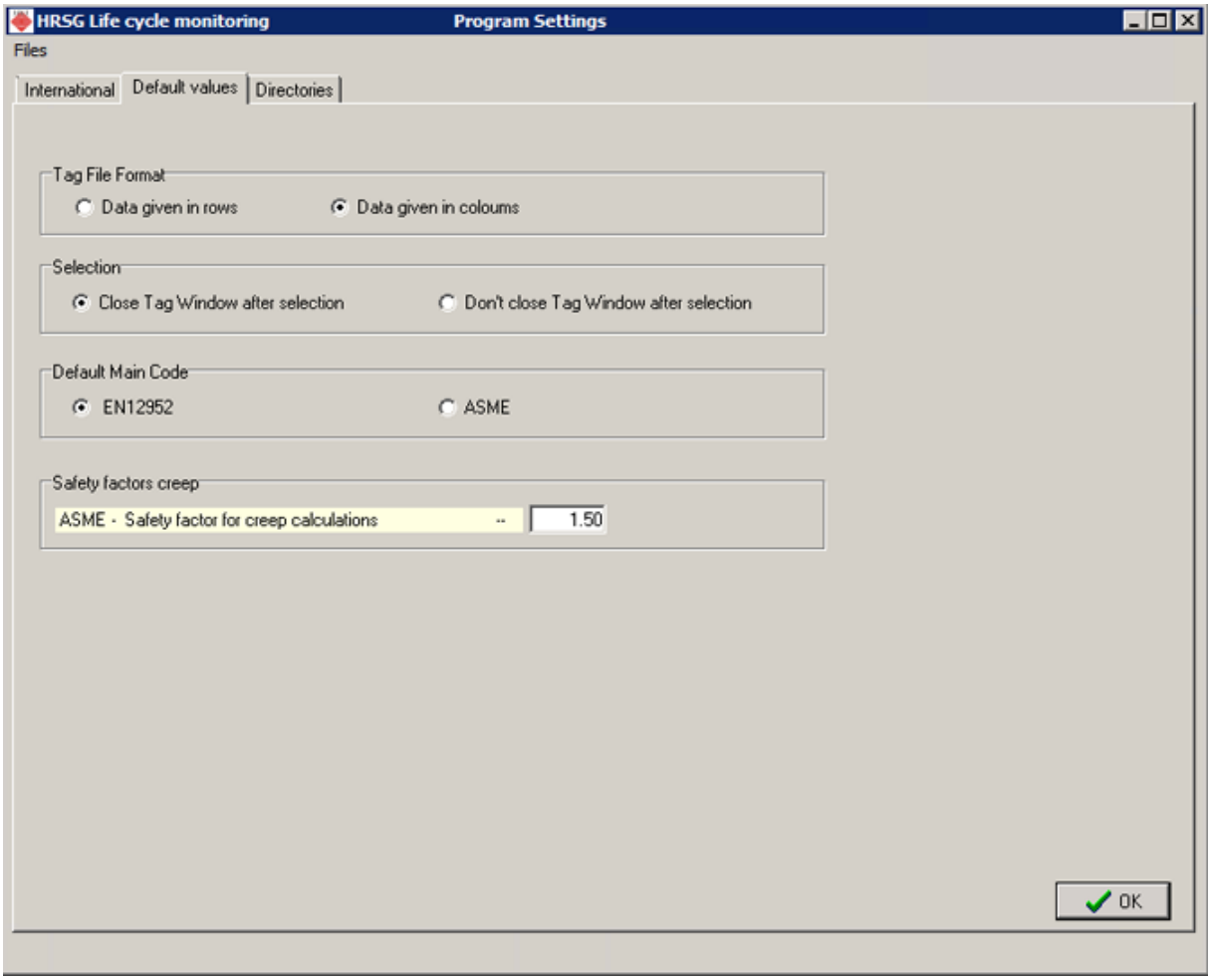
1. Click **OK** to immediately activate the settings.
2. From the **File** menu, select **Save Setting** to save the settings. The settings are saved as default values for the next time the software runs.

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3.5.2 Default Values Tab

The Default Values tab can only be accessed in Full or Partial Access modes.

The Default Values tab enables settings for data communication between the DCS, BSM software, and the calculation model for fatigue and creep damage.



Default Values screen

The following sections are displayed on the Default Values screen:

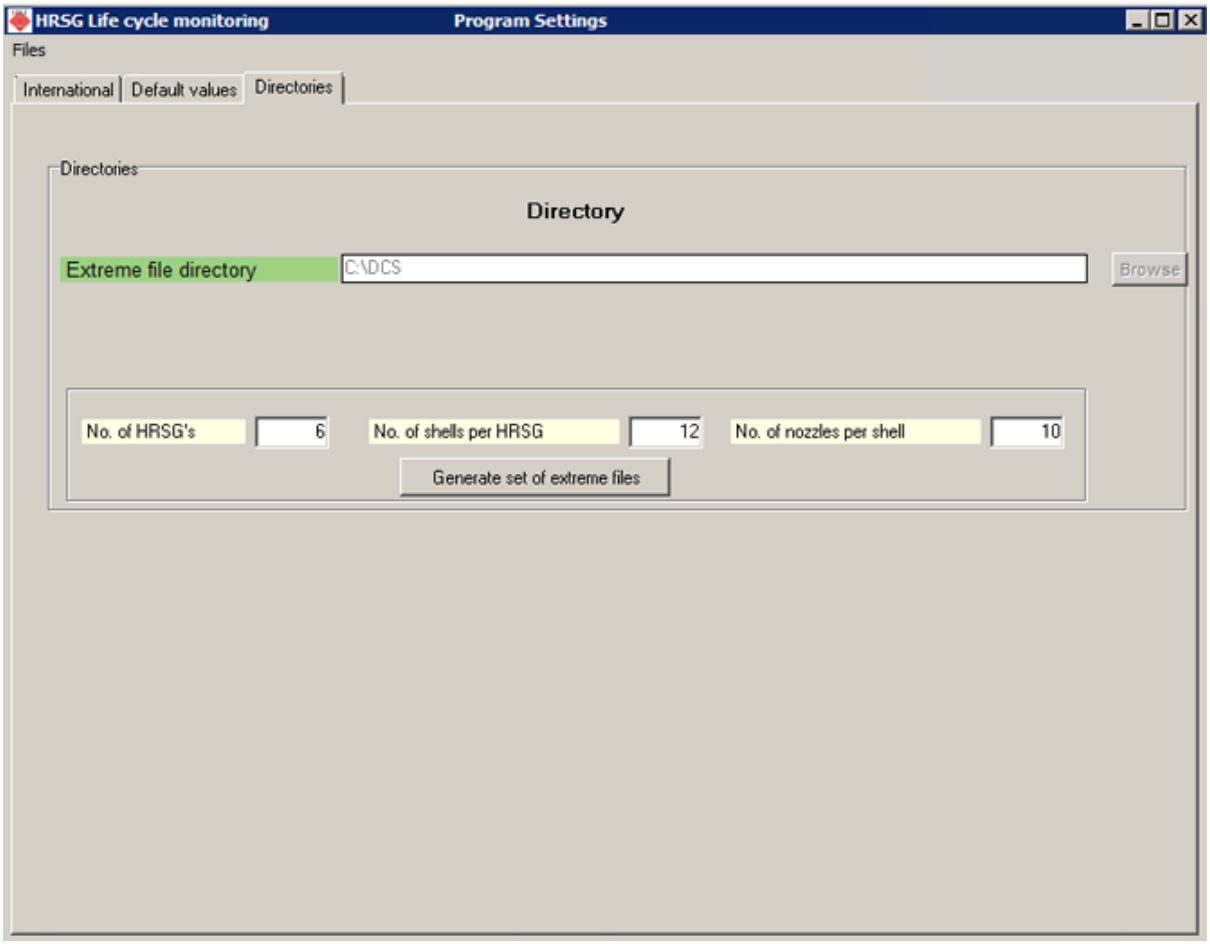
- **Tag File Format** specifies the file format for the .csv files used for data exchange between the DCS and BSM software.
- **Selection** determines the behavior of the user interface when the Show Tag Name page is selected. The Selection menu is not implemented in this release of the BSM software.
- **Default Main Code** determines the default calculation standard for fatigue and creep damage calculation. Currently, only EN12952 methods are implemented in this release of the BSM software.
- **ASME-Safety factor for creep calculation** specifies the safety factor to be used when deriving the material allowable stress values in the creep range when the time dependent creep rupture strength is known for use in the ASME analysis methodology. ASME methodology is not available in this release of the BSM software.

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3.5.3 Directories Tab

The Directories tab can only be accessed in Full or Partial Access modes.

The Directories tab defines a set of *extreme* files. Extreme files are text files that contain process and stress parameters for identified extreme points for every nozzle used in the cycle counting method. A set of extreme files must be declared *before* running the BSM software. The number and naming of the files depend on the given numbers of HRSGs, shells, and nozzles. The extreme files are used for troubleshooting during project setup and should not be modified by the user.



Directories Screen

- **To generate a set of extreme files:** from the directory specified under **Extreme file directory**, click **Generate set of extreme files**.

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The following values are written out for every detected extreme point (refer to EN12952-4):

- Steam pressure
- Steam temperature
- Inner wall temperature
- Middle wall temperature
- Mechanical stress value
- Thermal stress value
- Total stress value
- Extreme stress values for cycle detection
- Stress range for detected cycle
- Middle wall temperatures for operation point used in cycle detection.
- Fatigue damage factor for detected cycle
- Number of cycles detected with a stress range below the threshold for fatigue damage to occur

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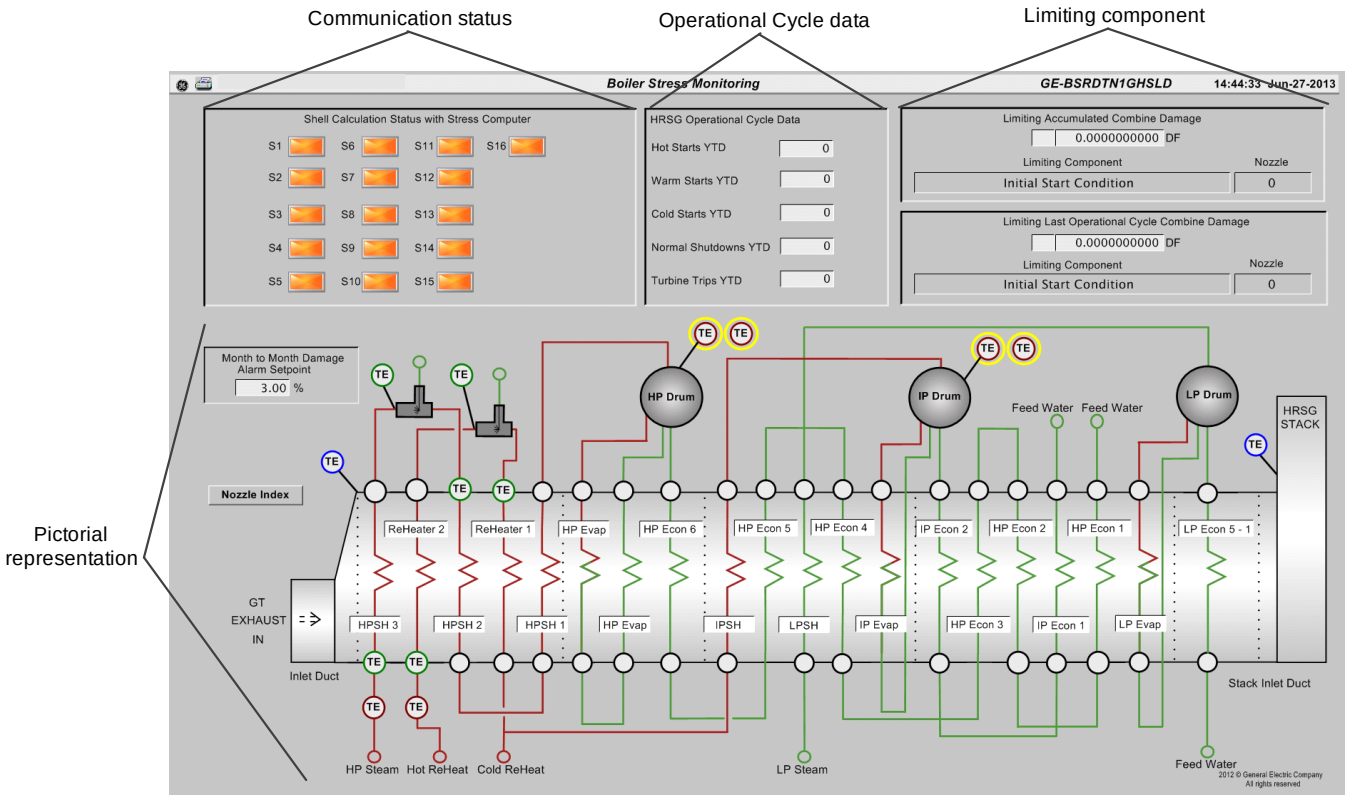
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4 HMI Screens and Operator Interface

Each HRSG has a dedicated BSM screen.

The BSM product requires no operator interaction to function. However, screens are provided for system monitoring. The operator interface screen consists of four zones:

- *Communication status*
- *Cumulative operational cycle data*
- *Limiting component* (contains identification and current values)
- *Pictorial representation with faceplates*



Block 1 - HRSG 11 HRSG Stress Screen

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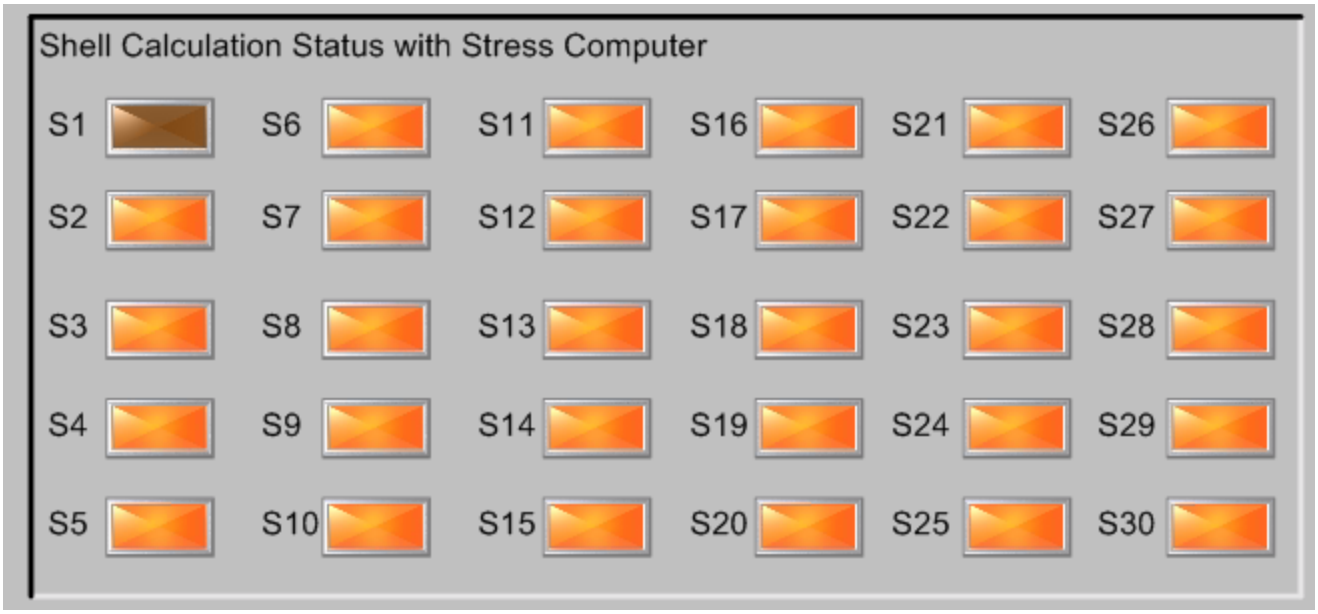
4.1 Communication Status

Communication Status indicates the communication health between the Mark VIe control system and the DXT online calculation engine. Communication is monitored at the controller level. Component communication is monitored individually to ensure that OPC-DA data exchanges are functioning properly and the calculation engine is available.

The DXT online calculation engine controls the data read/write and calculation cycle. The value is set prior to product shipment and is not normally field adjustable. The default time is 10 seconds between data read – execute calculations – write out results.

The variable, *FILE_EX*, enables communication monitoring at controller level. If *FILE_EX* does not increment due to communications loss and is not restored after four minutes, a loss of communication alarm will be generated for that component. If communications are lost to all components of a HRSG, an alarm is generated for the HRSG and the individual component alarms will be masked. A loss of communication alarm to all components could indicate the following:

- DXT online software has either crashed or locked up and requires restarting
- WorkstationST services have failed and require restarting on the BSM1 workstation
- BSM1 workstation computer has failed
- Possible UDH network problem

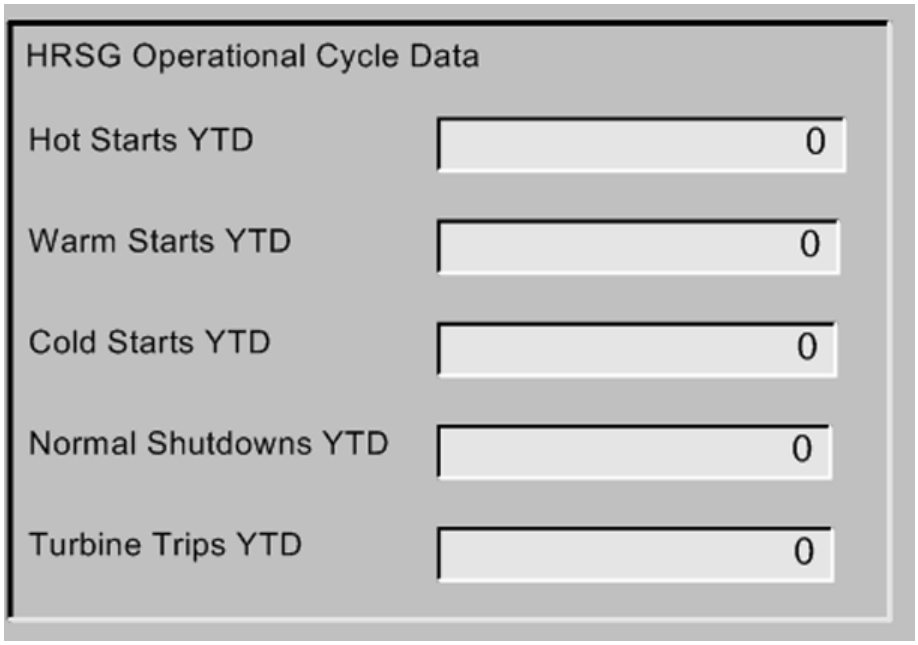


Communication Zone Indicating Shell 1 Not Communicating

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4.2 HRSG Operational Cycle Data

Operational Cycle Data provides a quick report of the starts and shutdowns of the HRSG gas turbine. HRSG counters are calendar-based and reset at the start of a new year. The counters give an intuitive feel for the stresses the HRSG is subjected to when compared with the limiting damage.



HRSG Operational Cycle Data

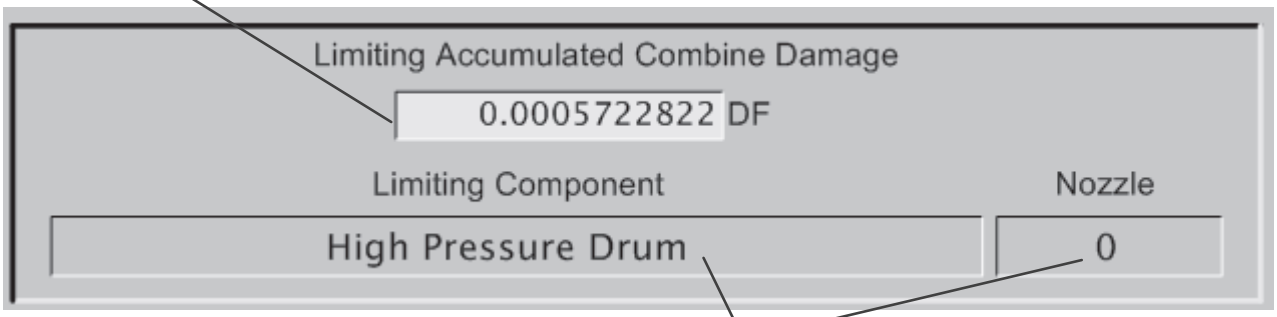
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4.3 Limiting Component

The Limiting Component consists of two primary pieces of data:

- **Limiting Accumulated Combine Damage** displays the specific HRSG component with the maximum amount of combined damage with its running accumulation of stress, fatigue and creep damage to date.
- **Limiting Last Operational Cycle Combine Damage** displays the component that saw the greatest accumulated damage during the last gas turbine cycle from start up to shut down and the accumulated combined damage during the cycle.

Accumulated Damage consists of stress, fatigue, and creep. Scale of 0 to 1.

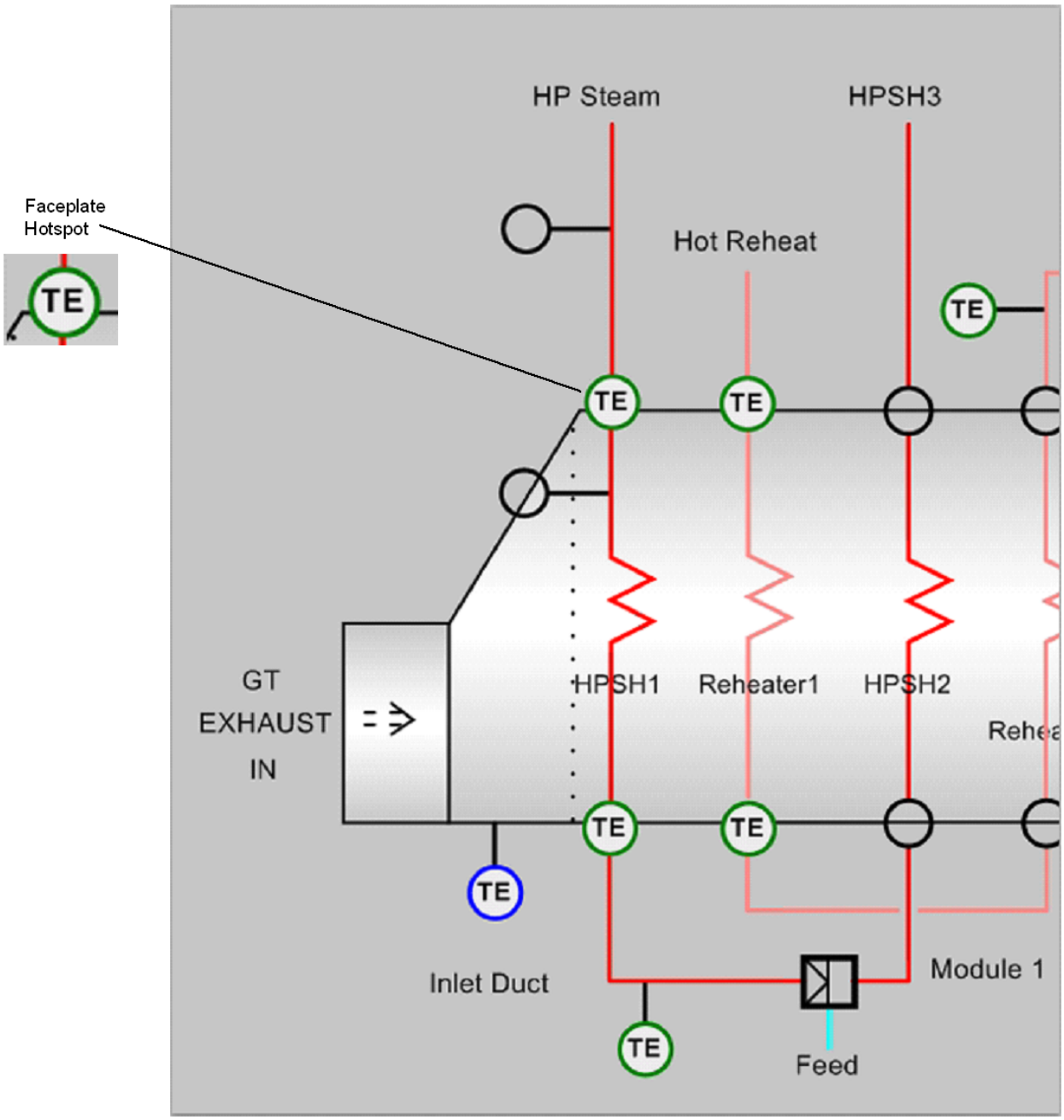


HRSG Component displays the greatest accumulated damage.

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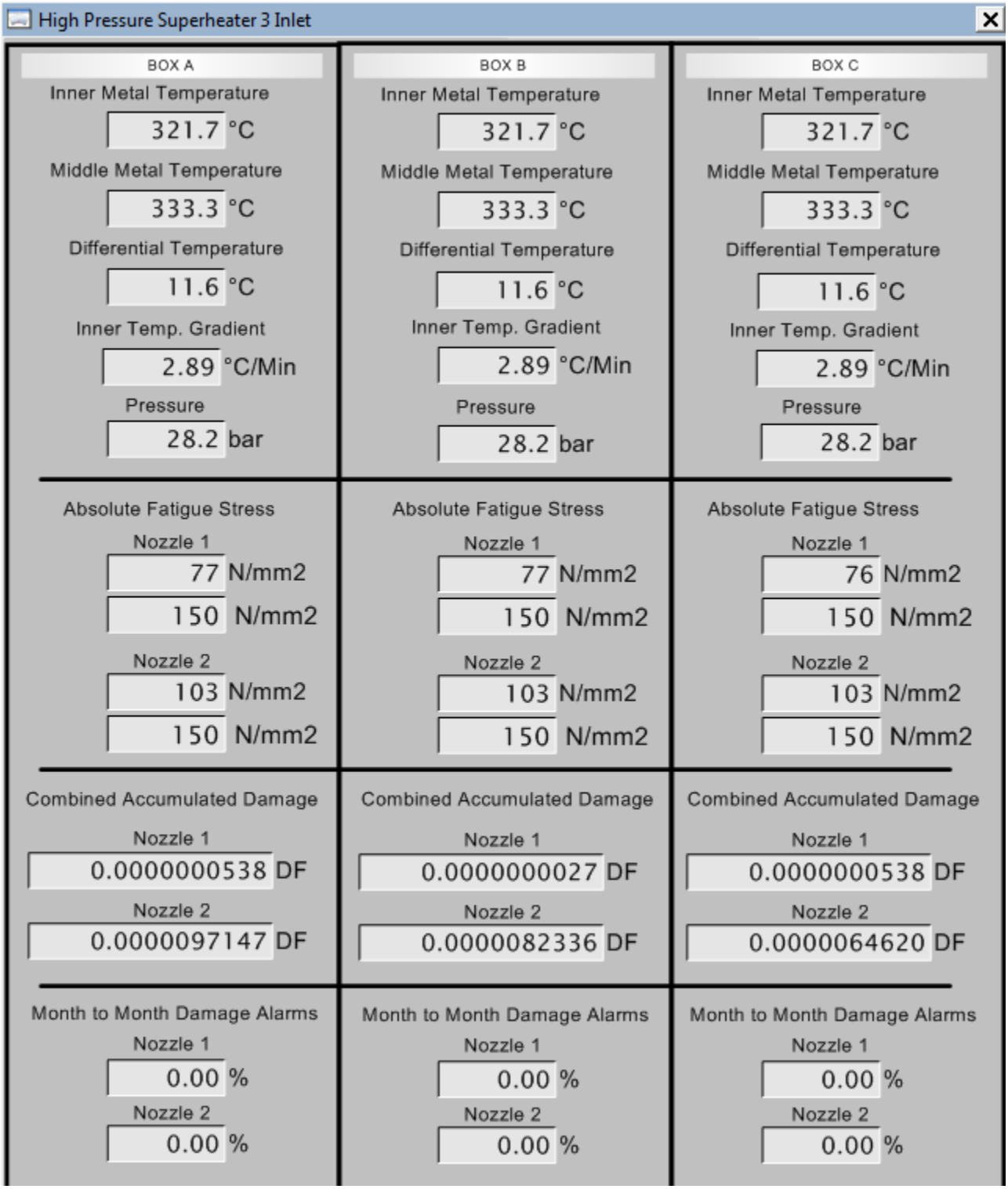
4.4 Faceplates

Each HRSG Stress graphic screen contains *hotspots* on the HRSG pictorial representation. Hotspots open faceplates that display critical data for each component, such as the process data containing temperatures, pressure, and accumulation for each monitored component nozzle.



HRSG Stress Graphic Screen With Hotspots

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Standard Faceplate Displaying HP Super-Heater

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A Nozzle Index faceplate is also provided for quick identification, which includes the nozzle number and the type of nozzle being monitored.

- **To access the Nozzle Index faceplate**: from the HRSG Stress graphic screen, select the **Nozzle Index** radio button.



Nozzle Index Faceplate

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5 Historical Data

Critical data for each monitored component is archived in the plant data historian, which includes the inner and mid-wall temperatures, drum or shell pressure, and the accumulated fatigue and creep for each nozzle of the shell. This data supports the user in developing trends for the accumulated shell damages, temperatures, and pressure to be overlaid with other plant process data, such as gas turbine exhaust gas temperature, steam flows, gas turbine watts, and so forth.

GE can provide consulting services to assist in the development of custom reports to meet site requirements.

Standardized reports can be created from the historian for operators' use in maintaining and evaluating plant condition. Since operational philosophy and metrics are not standard, GE does not provide any standard reports with the BSM system.

The BSM provides the following set naming convention for data archiving:

Convention 1 for process data:
{Controller Name}.B{block number}H{HRSG number}_S{shell number}_{descriptor}

For example:
B1H1.B1H1_S1_PRESS_60SEC is controller B1H1 - Block 1 - HRSG 1 - Shell 1 – PRESS_60SEC

Each tag ending in 60SEC indicates that the value of the tag has been processed through median or dual selects if the signal originates from either dual or triple transmitters, and the value has been passed through a 60 second averaging function.

Convention 2 for stress and creep values:
{Controller Name}.B{block number}H{HRSG number}_S{shell number}_N{nozzle number}_{descriptor}

For example:
B2H1.B2H1_S6_N2_CSADF_GE is controller B2H1 - Block 2 - HRSG 1 - Shell 6 – Nozzle 2 – Creep Stress Accumulated Damage Factor

Name	Description	Historian Deadband	Units
B1H1_S1_NZ1_CSADF_GE	B1H1_S1 Nozzle 1 Accumulated Creep Damage Factor GE	0	DF
B1H1_S1_NZ1_CADF_GE	B1H1_S1 Nozzle 1 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ1_FSADF_GE	B1H1_S1 Nozzle 1 Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_NZ2_CSADF_GE	B1H1_S1 Nozzle 2 Accumulated Creep Damage Factorr GE	0	DF
B1H1_S1_NZ2_CADF_GE	B1H1_S1 Nozzle 2 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ2_FSADF_GE	B1H1_S1 Nozzle 2 Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_NZ3_CSADF_GE	B1H1_S1 Nozzle 3 Accumulated Creep Damage Factor GE	0	DF
B1H1_S1_NZ3_CADF_GE	B1H1_S1 Nozzle 3 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ3_FSADF_GE	B1H1_S1 Nozzle 3Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_AW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1	DegC
B1H1_S1_AW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1	DegC
B1H1_S1_BW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP BELOW WATER 60 SECOND LAG	0.1	DegC
B1H1_S1_BW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP BELOW WATER 60 SECOND LAG	0.1	DegC
B1H1_S1_PRESS_60SEC	B1H1_S1 60 SECOND LAG PRESSURE INPUT	0.1	Bar

Typical Historian Points for a Shell (Component)

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6 Diagnostics and Alarms

Communications between the DCS and BSM program is monitored from each controller that transmits and receives data from the BSM. Each controller monitors communication for each HRSG component being monitored by the DXT online calculation engine.

Each HRSG controller monitors communications with the BSM online calculation engine for the assigned components. The BSM online calculation engine monitors for a change in the time stamp provided for each component by the DCS. When a time stamp changes, the BSM online calculation engine calculates and writes the output data to the csv2live.csv files, if used, or to the OPC DA file exchange variable, *FILE_EX*. A *FILE_EX* variable exists for each component. The value of *FILE_EX* changes every time the file is written. The controller monitors this tag for a change and, if a change is not detected for four minutes, declares communication is lost for that component.

The controller generates a diagnostic alarm that is sent to the Alarm Server and changes the communication display status for that component on the HRSG Stress CIMPLICITY* screen. If the controller detects a loss of communications for every component being monitored, it declares a communication loss to the online software and generates a diagnostic alarm, which displays on the HRSG Stress CIMPLICITY screen and sent to the Alarm Server. If communication to all components in a controller is lost within 30 seconds of each other, the individual diagnostic alarm is blocked and only the loss of communication diagnostic alarm is sent to the Alarm Server. If the condition exists for four hours or longer, the active diagnostic alarms are re-generated.

Each component has a collection of counters internal to its DCS controller logic. These counters increment when any diagnostic or process alarms are generated and should be reviewed for assistance in troubleshooting when higher than expected damage factors occur and no operational causes are suspected.

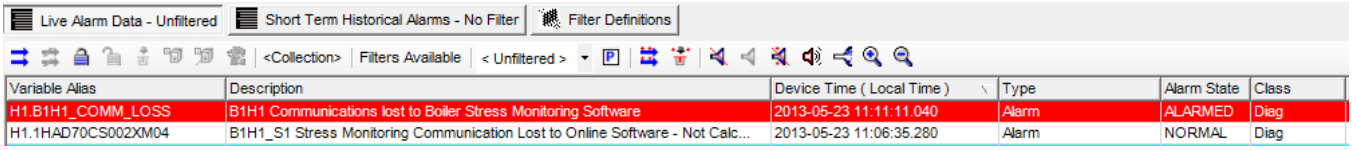
Typical faults that generate a Communications Loss alarm include:

- DXT online software failed
- BSM1 workstation hardware failed
- WorkstationST services failed on the BSM1 workstation
- Network communications lost between the workstation and the UDH network

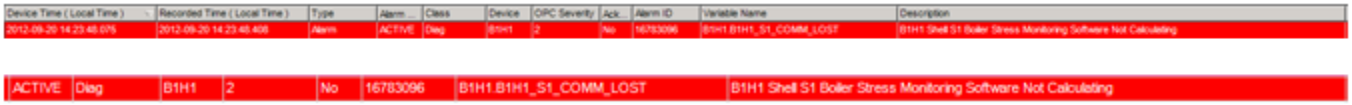
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Lost Communications Status on the HRSG Stress CIMPLICITY Screen



Diagnostic Communication Loss Alarming on the Alarm Server



Diagnostic Alarm for Loss of Communication for an Individual Component

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6.1 Process Alarms

The HRSG controller monitors the thermocouple temperatures for the inner and middle metal temperatures for each component. Each component being monitored by the BSM system has been configured with a metallurgically-based high alarm, a Hi-Hi alarm, which is at the upper range of the design temperature for the component. A Hi-Hi alarm is generated when the component exceeds either the maximum allowable temperature at which the component should not be operated beyond, or the 20 mA value to which the temperature transmitter has been calibrated. A low temperature alarm is also provided. The normal setpoint is 5°C (41° F) when the HRSG is in operation and decreases to 0°C (32° F) when not in operation.

If a thermocouple input exceeds any alarm setpoint threshold, a Level 3 process alarm is generated by the HRSG controller, and the appropriate suspend command is passed to the BSM online software. The DXT online calculation engine suspends calculations involving the affected temperature. The Hi-Hi alarm will be re-generated every four hours until the temperature signal drops at least 5°C (41 °F) below the maximum allowable temperature. The status field for the affected temperature is updated for that component on the BSM screen. Bad Quality diagnostic alarms are passed to the Alarm Server and the CIMPLICITY screen.

Device Time (Local Time)	Recorded Time (Local Time)	Type	Alarm	Class	Device	OPC Severity	Ack	Alarm ID	Variable Name	Description
2012-09-20 14:26:54.475	2012-09-20 14:26:54.487	Alarm	H	LVL_3	B1H1	2	No	67115249	B1H1.11HAD70CT002AXQ01_A	B1H1 HP Drum Above Water Inner Metal Temperature has Exceeded the Maximum Allowable Hi

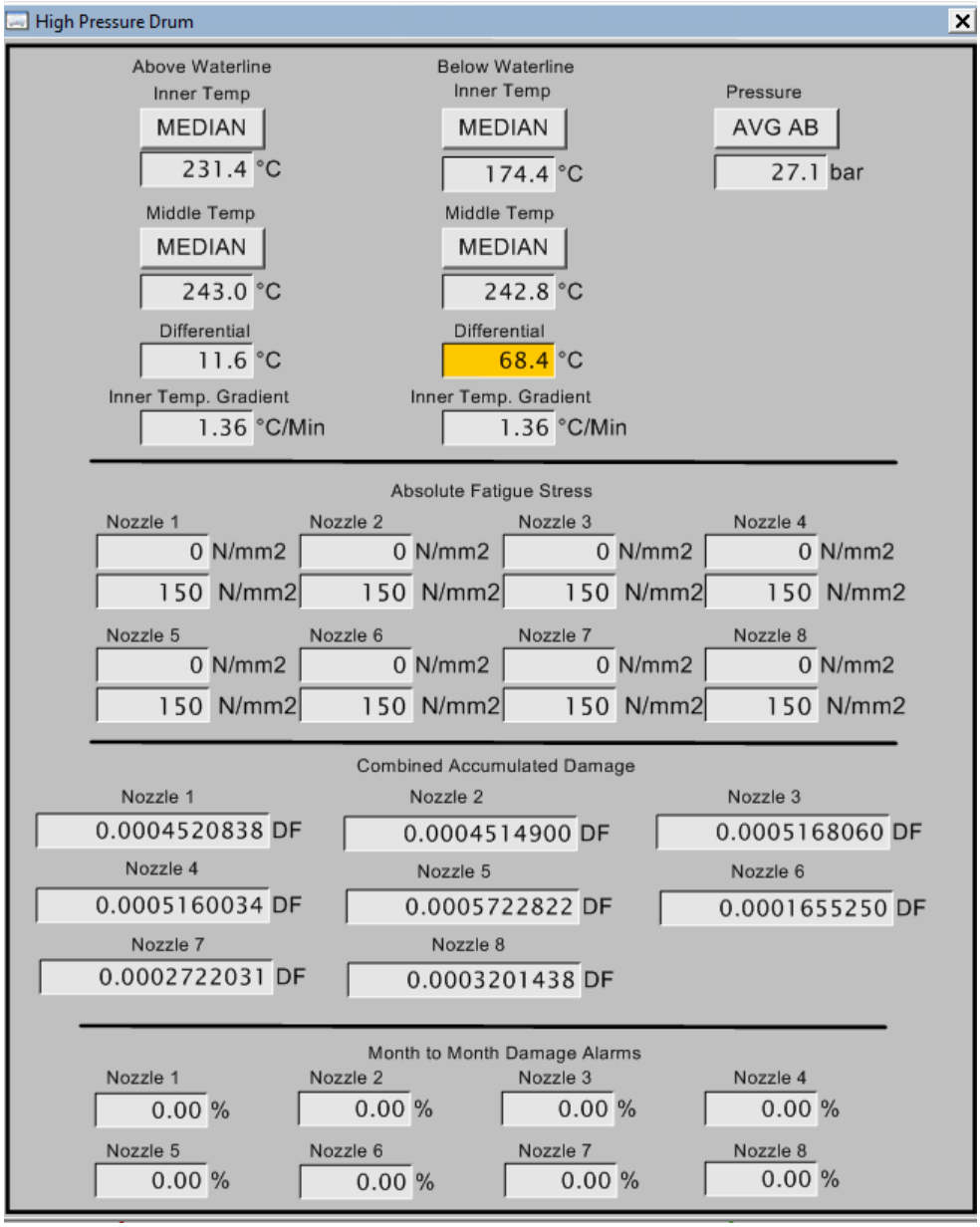
H	LVL_3	B1H1	2	No	67115249	B1H1.11HAD70CT002AXQ01_A	B1H1 HP Drum Above Water Inner Metal Temperature has Exceeded the Maximum Allowable Hi

High Temperature Alarm Sent to the Alarm Server

The following additional process alarms are provided through calculations:

- A Level 3 alarm is generated if the absolute value of the differential temperature between the inner and middle wall temperature of a component exceeds the expected maximum differential as calculated by the online software for that component. The differential temperature is calculated after signal filtering to ensure the alarm is not triggered by erroneous noise.

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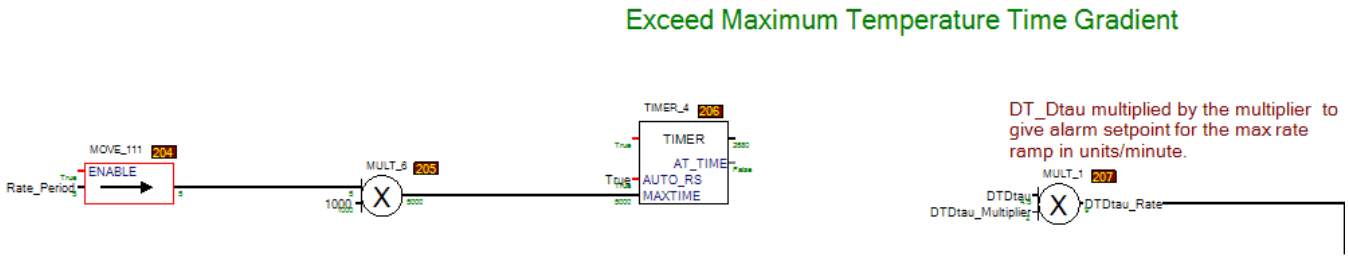
Typical Temperature Alarm Status Indication on BSM CIMPLICITY Screen

- A Level 3 alarm is generated when the inner metal temperature of a component exceeds the maximum temperature gradient (temperature ramp in degC/minute) and may indicate that the component is being ramped faster than was originally designed to yield the designed lifetime of the component.

The temperature gradient for the inner metal temperature is calculated and an alarm is generated when the temperature ramp rate in degC/minute exceeds the calculated gradient. The calculated value is generated as part of the calculations and configuration of the online calculation engine. The temperature gradient is calculated every 5 seconds, however, this time may be adjusted individually for each drum and shell by adjusting the local variable, *Rate_Period*, which is in seconds. Increasing the calculation time also increases the time between calculations, which may assist in troubleshooing spurious alarms.

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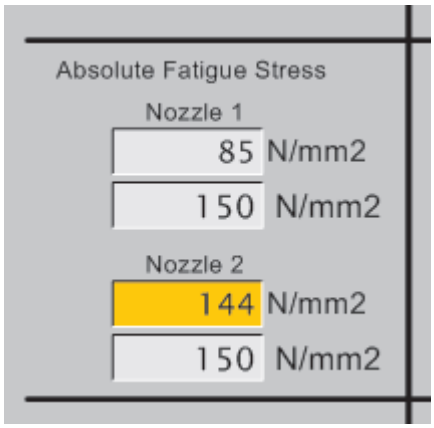
To raise or lower the setpoint threshold, each drum and shell has a multiplier value, *DTDtau_Multiplier*, to bias the calculated rate for the alarm setpoint. These constants can only be modified through controller logic accessed by using ToolboxST application. Once an adequate time base or setpoint multiplier value has been found, the initial value attribute for the variables should be changed to match to prevent losing the values during a controller reboot.



Typical Controller Logic with Variables Used to Adjust the Temperature Gradient Alarm

- A process alarm is generated when the absolute total fatigue stress per monitored nozzle of each component is exceeded. The high setpoint is adjustable from the associated component faceplate for each nozzle. This alarm alerts the operator of exceeding the total fatigue stress as determined by the HRSG vendor or plant engineers.

The set point is displayed beneath the actual stress value for each nozzle

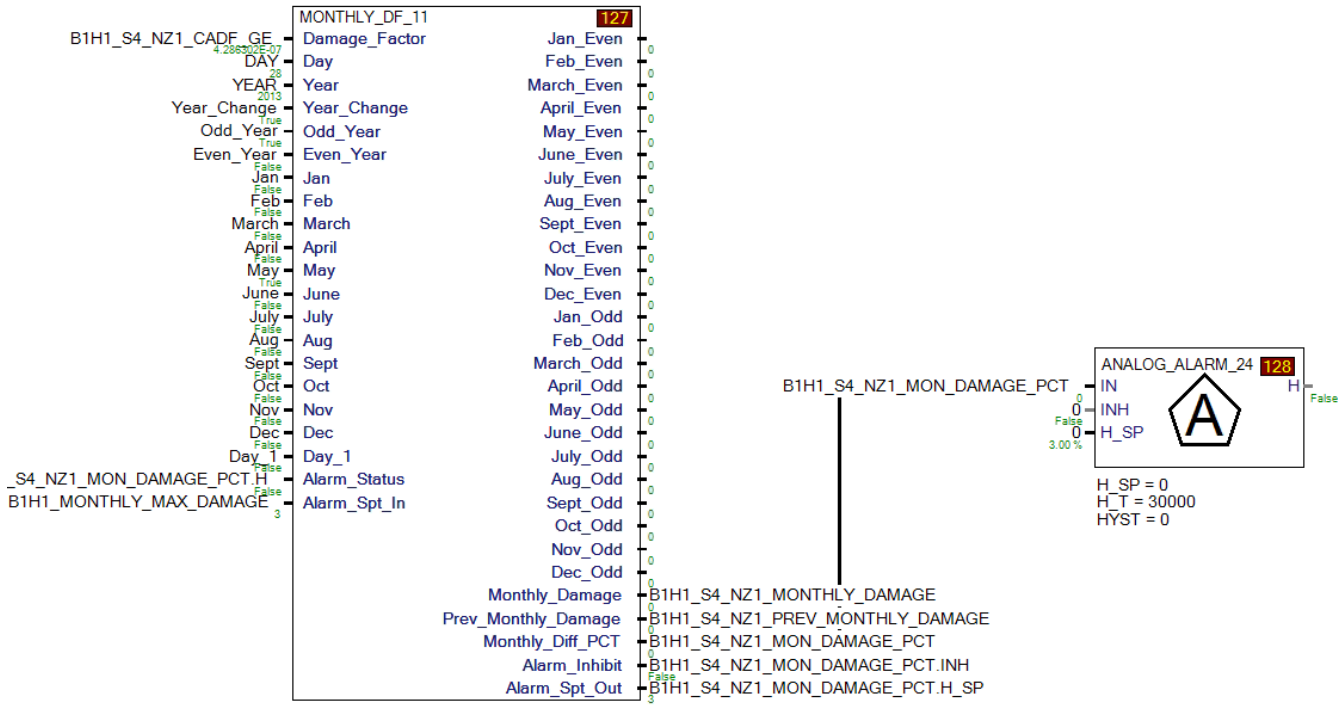


Absolute Fatigue Stress Faceplate

- A month-to-month combined damage factor alarm is provided for detecting unexpected accumulation of fatigue damage. Month-to-month combined damage is calculated for every nozzle of every monitored component on the HRSG. One master setpoint is used for all nozzles and is entered in percent. The month-to-month change is calculated at the start of a new month and this alarm is generated if the damage of the month just concluded is greater than the month previous to that by the percent setpoint.

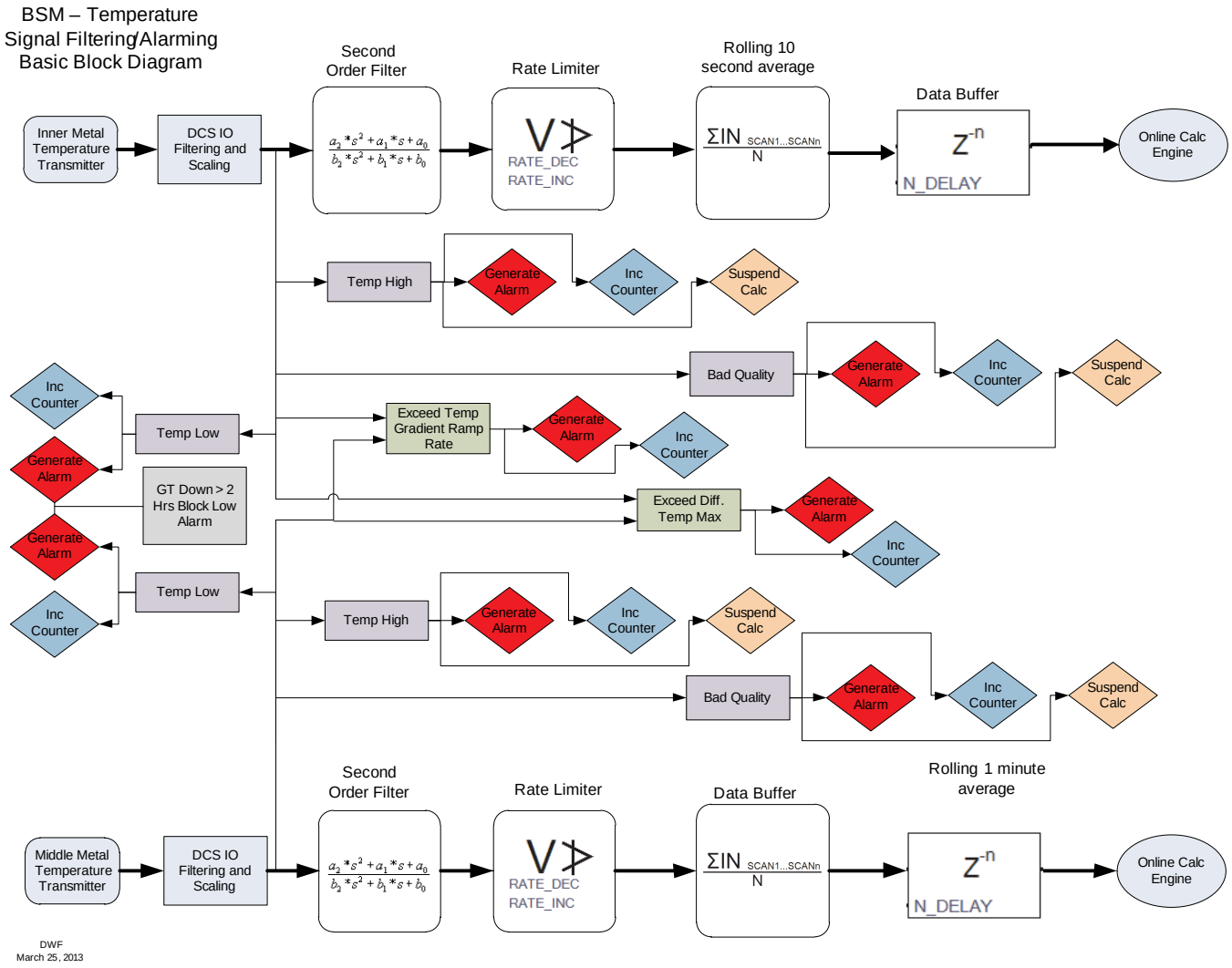
  The last month-to-month calculated value is displayed per nozzle of each component on the associated screen faceplate. However, the calculation is contained within a macro block in the Boiler Stress Logic of the HRSG controller and the macro block is designed to retain up to 24 months of previous comparisons broken down as twelve months of even year months and twelve months of odd year values. These values are not stored in non-volatile memory and a controller reboot will set them back to zero.

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Typical Macro Block for Month-to-Month Calculation Damage Factor

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Block Diagram of Process Signal Processing

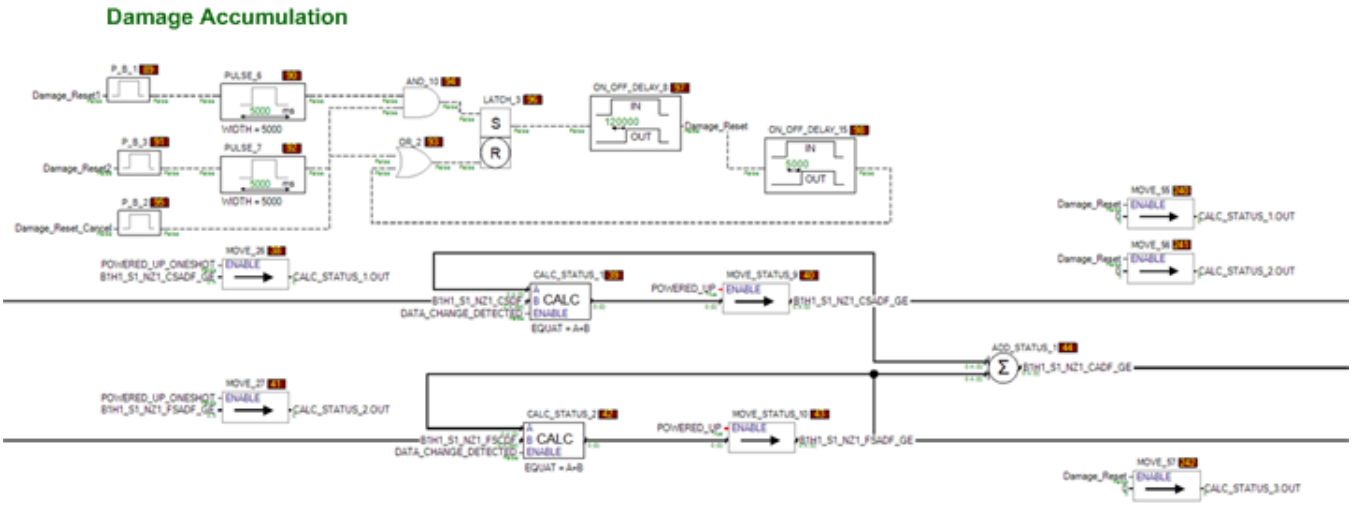
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6.2 Component Damage Factors

The online calculation engine calculates a Creep damage factor and a Fatigue damage factor fraction for each critical nozzle of a component. The Creep damage is calculated from successive data updates which occurs every 10 seconds. The Creep damage number is added in the Mark VIe DCS to the previous total to maintain a cumulative Creep damage value. The Fatigue damage number is updated when the online calculation engine detects a completed fatigue cycle and updates the calculated Fatigue damage for the cycle to the Mark VIe DCS. These two values are stored in non-volatile ram (NOVRAM) of the HRSG controller to ensure protection of the numbers during a controller reboot. The two numbers are summed together to give the Combined Accumulated Damage Factor. This value is scaled 0 – 1 DF.

Proper access privileges are required to reset the Creep and Fatigue non-volatile numbers and modify data in the controller.

A reset mechanism is in place in the controller logic to reset the non-volatile Creep and Fatigue damage numbers as needed, such as in a replacement of a full component. The reset only affects the non-volatile number in the controller with the history of the values preserved in the historian. The toggling of two internal push-buttons within 5 seconds of each other sets a latch and starts a two minute timer. A reset cancel internal push-button is also provided to cancel the reset before the two minute timer times out, as needed. Once the numbers have been reset, the only mechanism to restore the previous numbers is to manually enter them into the Creep and Fatigue non-volatile online value based on the historical value.



Typical Reset Logic for Non-Volatile Creep and Fatigue Variables

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